Replay–Plasticity Co-Design at 16 KiB for Patient-Specific Adaptation of a Tiny ECG Transformer

A Software Study Under an Analytical Persistent-State Budget

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Abstract

Patient-specific adaptation of tiny electrocardiogram (ECG) classifiers must fit replay exemplars, trainable parameters, gradients, and optimizer state into one small reserved adaptation budget. We study that allocation in a 6,643-parameter ECG transformer under an analytical 16 KiB constraint on *persistent state*—what a device must retain between adaptation steps, explicitly excluding peak training SRAM. The architecture exhibits a *replay-point cliff*: a 198-sample beat expands to 1,056 token elements and an 8,448-element feed-forward tensor while global pooling contracts it to 16, so only post-pool replay is storage-compressive, making maximum-volume post-pool replay the naive budget-optimal choice. Over all 21 patient-disjoint MIT–BIH DS2 records and five seeds (630 chronological adaptation cells), with metric, seed count, tests, effect size, and multiplicity family fixed before the final runs, a rank-1 encoder low-rank adapter with tens of raw replay samples raises per-patient macro-F1 from 0.666 to 0.803 and a rank-2 adapter to 0.809, each improving 17 of 21 patients against 12 for maximum head-only replay, with stronger pooled minority-class performance. Both beat the frozen model (Holm $p \leq 0.008$), but their *direct* comparison with maximum head-only replay is **not** significant ($p = 0.25$, 0.11), so maximum-volume replay is not established as uniquely optimal. Paired ablations support a contribution from encoder plasticity ($p = 0.005$) but not from replay volume. The evidence supports a **replay–plasticity allocation frontier** with several competitive operating points rather than a universal winner. This study is software-only: no hardware runtime, SRAM, latency, or energy is measured.

Keywords: arrhythmia classification; continual learning; electrocardiography; low-rank adaptation; patient-specific learning; replay; TinyML; transformers.

1 Introduction

Wearable and portable ECG systems collect long-duration cardiac signals outside the clinic. A static classifier trained on population data can still perform unevenly across patients, because beat morphology, rhythm prevalence, electrode placement, and acquisition conditions vary substantially between individuals. The problem is most acute for minority arrhythmia classes, whose prevalence is small relative to normal beats and whose morphology is patient-specific.

Busia *et al.* proposed a 6,643-parameter transformer for low-power arrhythmia classification on microcontrollers [1]. Such a model is attractive for embedded inference, but deployment-time personalization introduces a distinct resource problem. Continual adaptation on the device requires *persistent* state: trainable parameters, their gradients, optimizer moments, replay exemplars, replay labels, and buffer metadata. Under a microcontroller-motivated memory ceiling, these components cannot be designed independently, because every byte spent on one is a byte unavailable to the others.

Latent replay reduces storage by preserving intermediate representations instead of raw samples [5, 6]. In many convolutional networks, feature-map size decreases with depth, so replaying deeper features gives a relatively smooth memory–plasticity trade-off. The evaluated ECG transformer behaves differently. Tokenization *expands* the signal before pooling collapses it, so the storage cost of a replay exemplar is non-monotone in depth. We call this the *replay-point cliff*: only the post-pooling representation is smaller than the raw ECG window, and every intermediate tap is larger.

The cliff makes an intuitive design choice look optimal: because the post-pool vector is $\approx 11\times$ smaller than the raw beat, one can store hundreds of post-pool exemplars and adapt only a small classifier head. That intuition is arithmetically correct about *where* replay is cheap, but it does not settle *how* the budget should be spent. Retaining a small amount of encoder plasticity through low-rank adaptation, paired with only tens of raw replay samples, yields stronger pooled minority-class performance and more frequent positive patient-level changes than maximum-volume post-pool replay at the same ceiling. We do not claim it is statistically superior: maximum-volume replay is simply *not established as uniquely optimal*, and the budget admits several competitive allocations.

This paper frames the problem as **replay–plasticity co-design** and reports a pre-specified primary evaluation over the full eligible DS2 cohort. Relative to an earlier exploratory version restricted to an arrhythmia-informative panel and three seeds, every headline number here is computed over all eligible patient-disjoint DS2 records (21 after excluding record 202) with five seeds, with patient-bootstrap uncertainty, Holm-corrected paired tests, and effect sizes. The contributions are:

1. an architecture-level measurement of the replay-point cliff and its consequence for where replay may be stored (Section 3);
2. a complete persistent-state allocation formulation that jointly accounts for replay representation, trainable scope, gradients, and optimizer state, and a constrained configuration generator that turns a byte budget into a maximum replay count (Sections 3, 4);
3. a pre-specified patient-level comparison of frozen, head-only, post-pool adapter, and encoder-LoRA adaptation over all eligible DS2 patients with five seeds. The metric, seed count, paired tests, effect-size definition, and multiplicity family were fixed before the final runs. Record 202 was subsequently excluded following a subject-identity audit, and all affected analyses were rerun without changing the methods (Section 5);
4. paired ablations separating replay location, replay volume, and trainable depth, which support a plasticity contribution, do not support a direct replay contribution, and find no benefit from replay volume (Section 5);
5. robustness evidence on three axes: an Adam rank-1/rank-2 budget sweep showing that a reduced 12-record, three-seed Adam rank-1/rank-2 panel found no statistically detectable benefit from increasing the persistent-state budget from 8 to 64 KiB, adaptation from two of five trained alternative class-reweighted source checkpoints, and subject-level cross-database transfer to INCART and SVDB (Section 5);
6. a label-budget analysis quantifying how much supervision patient-specific adaptation actually requires (Section 5);

7. a memory-matched regularization baseline — an L2 source anchor on the B factor of the low-rank delta that costs zero persistent bytes — showing that replay provides substantially better source retention than this tested zero-byte B-factor anchor under the evaluated configuration (Section 5);

8. a leakage and reproducibility audit, and an explicit accounting of what the software-only evidence does and does not support—including the non-significance of the direct encoder-versus-maximum-replay comparison, the partial coverage of the budget grid, and the small subject count on INCART (Section 7).

We deliberately avoid claiming demonstrated microcontroller deployment. The 16 KiB ceiling is an *analytical, hardware-motivated persistent-state design constraint*: it covers what a device must retain *between* adaptation steps—trainable weights, gradients, optimizer moments, replay exemplars, replay labels, and buffer metadata. Peak training SRAM is instead dominated by activations (cf. TinyTL [22]) and is explicitly out of scope and unmeasured here. Runtime latency, energy, and nonvolatile-memory endurance are likewise outside the scope of this study and are not measured.

2 Related Work

2.1 Patient-specific ECG adaptation

Personalizing a heartbeat classifier to an individual is a long-standing idea. The mixture-of-experts classifier of Hu et al. combines a global model with a small patient-specific expert trained on a brief annotated segment [14], and Kiranyaz et al. train a compact 1-D convolutional network per patient from a few hundred labelled beats [15]. Both establish the central empirical fact we build on: a short labelled prefix of a patient's own record is worth a great deal, particularly for supraventricular beats that inter-patient models systematically miss. Neither, however, bounds the *state* that personalization costs. Their patient models are stored and trained without a byte budget, and neither considers what must be retained alongside the parameters — gradients, optimizer moments, and rehearsal data — which is precisely the quantity that decides whether personalization can run on a microcontroller. Large-scale ECG networks [18] sharpen the same gap in the opposite direction: accuracy at a parameter scale far beyond an embedded budget.

2.2 Inter-patient ECG classification

The MIT–BIH Arrhythmia Database remains a common benchmark for heartbeat classification [2, 3]. Patient-wise evaluation is substantially more demanding than random beat-level splitting, because morphology correlations within a record inflate optimistic estimates when beats from the same patient appear in both train and test. The de Chazal protocol partitions records into disjoint patient groups (DS1/DS2) and combines morphology with heartbeat-interval features [4]. We adopt the conventional DS1/DS2 record partition, correct its 201/202 cross-split subject overlap, and use the AAMI class mapping [20] but study *supervised adaptation* within an unseen patient's own chronological record, rather than a single static inter-patient model. Feature-selection work driven explicitly by cross-database generalization [16, 17] shows how fragile inter-patient performance is when the evaluation database changes, which motivates our external validation on INCART and the MIT–BIH Supraventricular Arrhythmia Database [19].

2.3 Latent replay under embedded constraints

Latent replay stores features extracted by a frozen prefix of a network [5]. Quantized latent replay has been demonstrated on TinyML platforms and motivates joint selection of replay precision and replay location [6]. On-device training within a few hundred kilobytes has also been shown for vision backbones [10]. These results largely concern convolutional architectures whose feature maps shrink with depth. Our replay-point cliff is architecture-specific: transformer tokens can be several times *larger* than the raw ECG window, so "replay deeper to save memory" does not hold.

Budget-aware replay with an explicit plasticity trade-off has recently appeared for vision continual learning [37], and memory–latency–accuracy trade-offs for latent replay have been characterized on extreme-edge RISC-V nodes [38] from the same lineage as our base model's platform. Our study differs in three ways: the replay-cost non-monotonicity is a property specific to an expand-then-pool ECG tokenizer rather than a general depth trend, the setting is patient-specific ECG adaptation rather than class-incremental vision, and the constraint is a byte-exact persistent-state budget rather than a bit-precision or feature-map allocation.

2.4 Source-free and test-time adaptation

A parallel line of work adapts a trained model to a shifted target distribution without labels, either by entropy minimization over test batches [23] or by source-hypothesis transfer [24]. These methods are attractive on a device because they need no annotation, but they adapt statistics and affine parameters rather than allocating a fixed memory budget, and they do not address the class-imbalance problem that dominates minority-class ECG performance. Our setting is deliberately the supervised one: we assume a short labelled prefix, which is the assumption under which the patient-specific ECG literature operates, and ask how to spend bytes given it.

2.5 Continual-learning regularization

Elastic weight consolidation penalizes changes to parameters estimated to be important for previous tasks [7]. Under our accounting, storing a full-model FP32 parameter anchor plus a Fisher value requires $6643 \times 8 = 53{,}144$ bytes, already above the nominal 16 KiB ceiling before any replay. The same arithmetic applies to synaptic intelligence [31]; distillation-based approaches such as learning without forgetting [28] avoid the anchor but require a copy of the source model's outputs, and gradient-projection methods [30] still keep an episodic memory. This establishes infeasibility only for *full-model* FP32 regularization under the stated assumptions; restricted or compressed regularization remains a valid future baseline and we do not claim that regularization is universally infeasible. Work on very small episodic memories [29] suggests that the replay side of this trade-off degrades gracefully, which is consistent with what we measure.

Regularization has also been brought inside the low-rank setting directly. Zheng et al. propose **EWC-LoRA** [40], which regularizes a *shared* low-rank update with a Fisher-based importance estimate, keeping storage and inference cost constant as tasks accumulate. That is the closest published method to the regularization arm of our study, and the comparison is worth stating precisely rather than by omission. Zheng et al. study Fisher-based regularization of shared low-rank updates in *task-sequential* vision and language continual learning. Our setting instead asks how replay and trainable depth should share a *kilobyte-scale persistent-state budget* for patient-specific ECG adaptation, where the binding constraint is bytes rather than task count. **We do not implement EWC-LoRA**, and therefore restrict our regularization comparison to the tested B-factor anchor (Section 5.4); we make no claim about how EWC-LoRA would perform under a 16 KiB ceiling, and note that a Fisher estimate is itself a persistent-state cost that would have to be charged against the budget.

2.6 Parameter-efficient adaptation

LoRA freezes the original weight matrix and learns a low-rank update [8]. Although introduced for large language models, the mechanism is equally useful when the goal is to preserve a tiny inference model while exposing only a few trainable parameters and their optimizer state. Transformer continual-learning methods such as DyTox reduce task-specific parameter growth [9], and bottleneck adapters achieve the same end by inserting small trainable modules into a frozen backbone [21]. On the systems side, TinyTL observes that trainable *parameter* count is the wrong memory target because activations dominate [22], and on-device training has been demonstrated within a few hundred kilobytes [10] on runtimes such as TensorFlow Lite Micro [33], whose workloads the MLPerf Tiny suite standardizes [32]. The problem has stayed active: recent TinyML continual-learning work adapts model *size* to the task and distils the retained set rather than storing raw exemplars [34], and a 2025 survey of replay-based continual learning on edge hardware confirms that the binding constraint in practice is how much history fits in memory rather than which replay rule is used [35] — which is the premise this paper takes to its conclusion by making the byte budget the independent variable. Parameter-efficient adaptation of ECG foundation models, including LoRA, is now surveyed as a standard branch [36], but at a parameter scale far above the microcontroller regime studied here. Our accounting is the persistent-state complement to that observation: we bound what must be *retained* between updates rather than what peaks during one. None of this work studies the ECG-specific replay-depth discontinuity under a kilobyte-scale payload budget.

2.7 Replay selection

Replay performance depends on which exemplars are retained. Herding, gradient-based selection, and bilevel coreset optimization are representative approaches [11–13]. Selection quality matters, but our evidence shows that a prior question must be answered first: *how much of the budget should be spent on replay at all*, versus on representation plasticity? We therefore hold the selection policy fixed and vary the allocation between replay volume and trainable depth.

2.8 Class imbalance in minority-beat classification

Supraventricular and fusion beats are rare, and the source model's failure on them is what patient adaptation must repair. Re-weighting by effective sample number [26] and focal loss [25] are the standard remedies; we use both to build the class-reweighted source variants of RQ5, in order to test whether our adaptation gain is merely compensating for a weak source rather than adding anything.

2.9 What is new here, and what is not

We want to be explicit about the boundary. Low-rank adaptation [8], replay [11], latent replay [5], patient-specific ECG classification [14, 15], and the tiny ECG transformer architecture itself [1, 27] are all established, and none of them is claimed as a contribution of this paper.

What is new is their *fixed-byte co-design*: measuring that replay cost is non-monotonic in depth for a tiny ECG transformer, formulating a single persistent-state budget that replay volume and trainable depth must share, generating budget-matched configurations from that constraint, and evaluating the resulting allocation under controlled patient-sequential adaptation. To our knowledge the allocation question — given a fixed number of bytes, how much should be spent on remembering the source versus on the freedom to change the representation — has not been posed for this class of model. Table 1 places the present work against the closest prior studies.

Table 1: Positioning against representative prior work. "Budget" is an explicit byte constraint on retained adaptation state, not a parameter count.

Work	ECG	Pt.-spec.	Replay	Trainable scope	Byte budget
Hu [14]	yes	yes	—	patient expert	no
de Chazal [4]	yes	no	—	full model	no
Kiranyaz [15]	yes	yes	—	full 1-D CNN	no
Hannun [18]	yes	no	—	full model	no
Pellegrini [5]	no	no	latent	partial net	no
Ravaglia [6]	no	no	quant. latent	partial net	yes
Lin [10]	no	no	—	sparse layers	peak
Hu [8]	no	no	—	low-rank	no
Busia [1]	yes	no	—	inference only	infer.
This work	yes	yes	raw + post-pool	head / adapter / LoRA	**persist.**

3 Problem Formulation and Method

3.1 Patient-sequential adaptation

Let $\mathcal{D}_0$ denote the source-patient (DS1) training data and f_{θ_0} the source model. For an unseen target patient p, an early chronological segment $\mathcal{D}_p^{\text{adapt}}$ supplies labeled beats for supervised adaptation, while a later segment $\mathcal{D}_p^{\text{test}}$ is held out for evaluation: no test beat and no test label takes part in adaptation or model selection. (The released preprocessing filters each record as a whole, so the test *signal* is not strictly unseen near the boundary; Section 4 bounds that dependency and Section 5.7 reruns the experiment without it.) Each patient begins from the *same* source checkpoint θ_0; there is no cross-patient state carry-over. A frozen replay memory $\mathcal{R}_0$ of source-domain exemplars is mixed into adaptation to limit forgetting.

For a metric $\mathcal{M}$, the per-patient target gain and the per-patient *source-domain change* are

$$G_p = \mathcal{M}\big(f_{\theta_p}, \mathcal{D}_p^{\text{test}}\big) - \mathcal{M}\big(f_{\theta_0}, \mathcal{D}_p^{\text{test}}\big)\,, \qquad \Delta_p^{\text{src}} = \mathcal{M}\big(f_{\theta_p}, \mathcal{D}_{\text{test}}^{\text{src}}\big) - \mathcal{M}\big(f_{\theta_0}, \mathcal{D}_{\text{test}}^{\text{src}}\big)\,. \tag{1}$$

Both are *signed changes measured in the same direction*: positive means the adapted model is better than the frozen source on that domain, negative means worse. We therefore report Δ_p^{src} throughout and call it the **source-domain macro-F1 change**, not "forgetting". Every source-domain number in this paper is negative, meaning source performance fell by that amount; a value of -0.027 is a smaller loss than -0.136. We avoid the word forgetting for the quantity itself because the two common conventions differ in sign and mixing them is a standard source of confusion. The primary unit of analysis is the patient: $\mathcal{M}$ is per-patient macro-F1 over the AAMI classes present in that patient's test segment. Fig. 1 summarizes the adaptation protocol end to end.

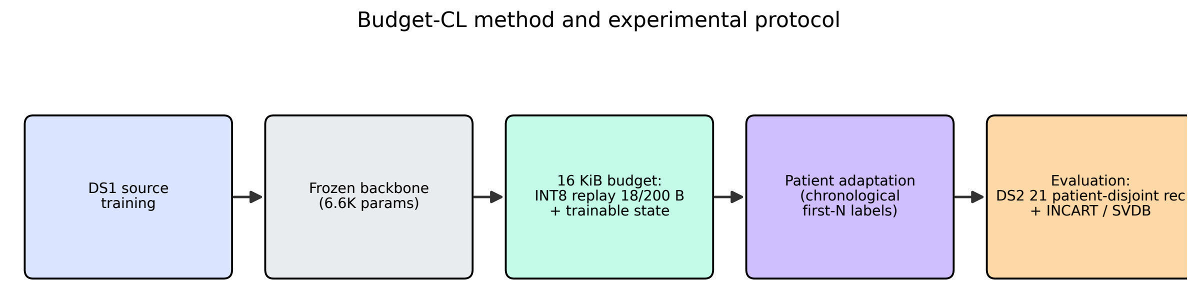

Figure 1: Method and experimental protocol. A DS1-trained tiny ECG transformer is adapted per patient: source replay is mixed with the early (chronologically first) labeled patient segment, while the later segment is held out for testing: it contributes no beat, label, normalization statistic, early-stopping signal, or model-selection information. The bounded whole-record filter dependency near the chronological boundary is quantified and removed in the split-first sensitivity analysis of Section 5.7. Adaptation may touch the classifier head, a post-pool residual adapter, or an encoder-attention LoRA update, each drawing from the same persistent-state budget. The budget covers persistent adaptation state; peak temporary training memory and hardware runtime are not evaluated.

3.2 Analytical adaptation-state model

What "persistent state" means here. We use *persistent adaptation state* to mean **a statically reserved writable region that the adaptation procedure needs across update steps**. It does *not* imply that every included quantity is nonvolatile or must survive power loss, and it is *not* total SRAM. The budget covers trainable weights, optimizer moments, replay values, replay labels, and buffer metadata, all of which must persist between updates. It also charges gradients, which are strictly update-time working state rather than retained state; we include them because an implementation that materializes a full gradient vector must reserve that region, and excluding them would understate what a device has to set aside. It excludes forward and backward activations, stack and workspace, and allocator overhead, which are transient peak-SRAM quantities and are not measured anywhere in this paper (Table 25).

A complete persistent-state accounting includes

$$B_{\text{persistent}} = B_{\text{weights}} + B_{\text{gradients}} + B_{\text{optimizer}} + B_{\text{replay}} + B_{\text{labels}} + B_{\text{metadata}} + B_{\text{padding}}. \tag{2}$$

The experiments use FP32 trainable weights, FP32 gradients, and two FP32 Adam moments, giving a per-parameter trainable cost of

$$b_{\text{train}} = \underbrace{4}_{\text{weight}} + \underbrace{4}_{\text{grad}} + \underbrace{8}_{\text{2 moments}} = 16 \text{ bytes/parameter}. \tag{3}$$

A stored replay item costs more than the tensor it carries, and conflating the two is an easy way to understate a budget. We therefore separate the *value payload* from the *serialized record*:

$$b_l^{\text{record}} = b_l^{\text{payload}} + b_{\text{label}} + b_{\text{flags}} + b_{\text{pad}}, \tag{4}$$

where b_l^{payload} is the INT8 tensor at replay location l and the remaining terms are the per-entry label, the validity and class-id flags, and any alignment padding. Concretely $b^{\text{payload}} = 18\,\text{B} \to b^{\text{record}} = 21\,\text{B}$ for post-pool and $200\,\text{B} \to 203\,\text{B}$ for raw, at 1-byte alignment. For P trainable parameters and N replay items the reserved adaptation state is

$$B(P, N, l) = 16P + N\, b_l^{\text{record}} + B_{\text{bufmeta}}, \tag{5}$$

with B_{bufmeta} the fixed per-buffer overhead (write index, count, reservoir state, class counters, and quantization parameters: 49 B post-pool, 54 B raw). The constrained replay-count generator that fixes a fair budget match is therefore

$$N_{\max}(P,l) = \left\lfloor \frac{B_{\text{budget}} - 16P - B_{\text{bufmeta}}}{b_l^{\text{record}}} \right\rfloor . \tag{6}$$

Every byte total in this paper uses b_l^{record}, never b_l^{payload}; where the text quotes 18 or 200 bytes it is naming the payload explicitly. Every configuration in this paper is generated from Eq. (6) so that $B(P, N_{\max}, l) \leq 16{,}384$ B holds by construction; the packed byte total for each arm is asserted per adaptation cell. Table 3 lists the resulting trainable counts, replay counts, and byte totals.

One-beat decision latency from the post-RR feature. The model consumes both the preceding and the *following* RR interval. The following interval is only known once the next R peak has been detected, so a classification cannot be emitted at the current R peak: **the task evaluated here is delayed heartbeat classification, one beat behind the signal**, not zero-latency classification. This is a property of the source architecture we adopt rather than of the adaptation method, and it applies equally to every arm including the frozen baseline, so it does not affect any comparison in this paper. It does bound the deployment claim, and comparing pre-RR-only, predicted-post-RR, and delayed variants is left as future work.

3.3 Replay-point cliff

The base model accepts a 198-sample ECG window and two RR-derived interval values. A convolutional embedding produces 66 tokens of width 16; the feed-forward hidden dimension is 128; mean pooling then contracts the sequence, and the two RR values are fused post-pool. Table 2 and Fig. 2 give the per-item INT8 payload at each candidate replay tap. Only the post-pool (and post-pool+RR) representation is smaller than the raw input; convolution tokens are 5.3× and the FFN hidden tensor 42.7× larger. The cliff is deterministic for this architecture and fixes *where* replay may be stored, but not *where it should be.*

Table 2: Replay-point sizes and the *serialized* INT8 cost of one stored exemplar at each candidate tap. "Values" is the tensor itself; "+RR" adds the two RR-interval features the model also consumes, without which an exemplar cannot be replayed; "+meta" adds the per-entry label, valid flag, and class id. The stride is the per-entry cost under 1-byte alignment, which is exact here because every field is a byte. Only the post-pool tap is storage-compressive.

Replay point	Shape	Values	+RR	+meta	× raw	Verdict
Raw ECG window	198 × 1	198	200	203	1.00	baseline
After conv embedding	66 × 16	1,056	1,058	1,061	5.33	larger (disqualified)
Encoder output	66 × 16	1,056	1,058	1,061	5.33	larger (disqualified)
FFN hidden	66 × 128	8,448	8,450	8,453	42.67	larger (disqualified)
Post-pool (mean)	16	16	18	21	0.08	**smaller (viable)**
Post-pool + RR concat	18	18	18	21	0.09	**smaller (viable)**

Buffer-level fixed metadata is charged once, not per entry: a 4 B write index, a 4 B count, 16 B of reservoir state, 5 × 4 B class counters, and a 4 B scale plus 1 B zero point per quantized tensor. That is 49 B for a post-pool buffer (one quantized tensor) and 54 B for a raw buffer (two: the beat and the RR pair). Under 4-byte alignment the strides become 24 B and 204 B; we report the 1-byte figures and account for padding explicitly.

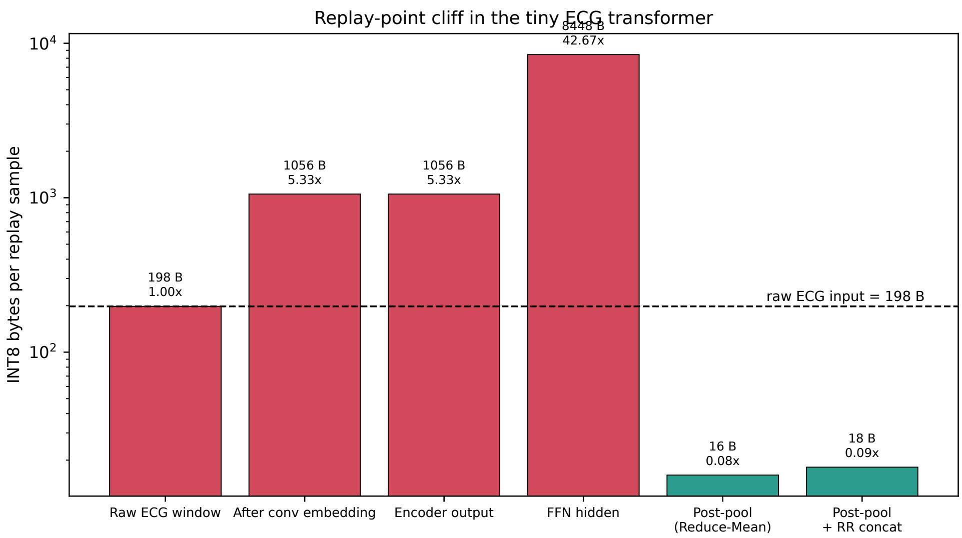

Figure 2: Replay-point cliff (log scale). Token and feed-forward representations expand relative to the raw ECG window; only post-pool replay is storage-compressive. Storing post-pool exemplars therefore maximizes replay *count*, which motivates—but does not justify—maximum-volume post-pool replay.

3.4 Adaptation arms

Six arms span the replay-volume / plasticity trade-off at 16 KiB:

- **A0** — frozen source model (reference; 0 trainable, 0 replay);
- **A1** — 95-parameter linear head, maximum post-pool replay;
- **A2** — rank-1 post-pool residual adapter plus head, post-pool replay;
- **A3** — rank-2 post-pool residual adapter plus head, post-pool replay;
- **A4** — rank-1 encoder-attention LoRA plus head, raw replay;
- **A5** — rank-2 encoder-attention LoRA plus head, raw replay.

For fused representation $z \in \mathbb{R}^{18}$, the post-pool adapter applies $z' = z + BAz$ with $A \in \mathbb{R}^{r\times 18}$, $B \in \mathbb{R}^{18\times r}$. For an encoder projection W, LoRA applies $W' = W + BA$ with $A \in \mathbb{R}^{r\times d_{\text{in}}}$, $B \in \mathbb{R}^{d_{\text{out}}\times r}$. A1–A3 spend the budget on *volume*: 650–705 post-pool exemplars at 18 B of payload each (21 B serialized). A4–A5 spend it on *depth*: only 52–62 raw exemplars at 200 B payload (203 B serialized) each, but expose the encoder to low-rank updates. For context, full-model Adam-style trainable state would need $6643 \times 16 = 106{,}288$ bytes, far beyond the ceiling, so full fine-tuning is infeasible under the constraint. Fig. 3 shows how each arm partitions the same 16 KiB.

Table 3: Complete arm definitions. "Trainable tensors" names exactly what receives gradients; every other weight is frozen. Persistent bytes cover trainable weights, gradients, two Adam moments per parameter, INT8 replay values, per-entry label/valid/class-id metadata, buffer-level metadata, and alignment padding. Replay counts are the maximum that fit, generated by the constrained configuration generator. The right-hand block repeats the generation against an effective ceiling of 15,360 B, i.e. a 1 KiB firmware implementation reserve; every arm was re-run at those reduced counts (Section 5.6).

				16,384 B ceiling		+1 KiB reserve	
Arm	Replay	Trainable tensors	Params	Items	Bytes	Items	Bytes
A0	—	none (frozen)	0	0	0	0	0
A1	post-pool	head W, b	95	705	16,374	656	15,345
A2	post-pool	post-pool adapter B_1A_1 + head	131	678	16,383	629	15,354
A3	post-pool	post-pool adapter B_2A_2 + head	167	650	16,371	601	15,342
A4	raw	encoder LoRA r=1 on $W_{q,k,v,o}$ + head	223	62	16,208	57	15,193
A5	raw	encoder LoRA r=2 on $W_{q,k,v,o}$ + head	351	52	16,226	47	15,211
R1	—	encoder LoRA r=1 + head, L2 source anchor	223	0	3,568	0	3,568
R2	raw	encoder LoRA r=1 + head, L2 source anchor	223	62	16,208	57	15,193

All arms use Adam ($b_{\text{train}} = 16$ B/parameter: 4 weight, 4 gradient, 8 for two moments), FP32 weights, and INT8 replay. Replay and target beats are concatenated rather than mixed at a fixed per-batch ratio, so the effective replay fraction is $N_{\text{replay}}/(N_{\text{replay}} + N_{\text{target}})$ with $N_{\text{target}} = 500$. The R-arm source anchor adds *no* persistent bytes: the anchor is the frozen base weight, which is already stored for inference.

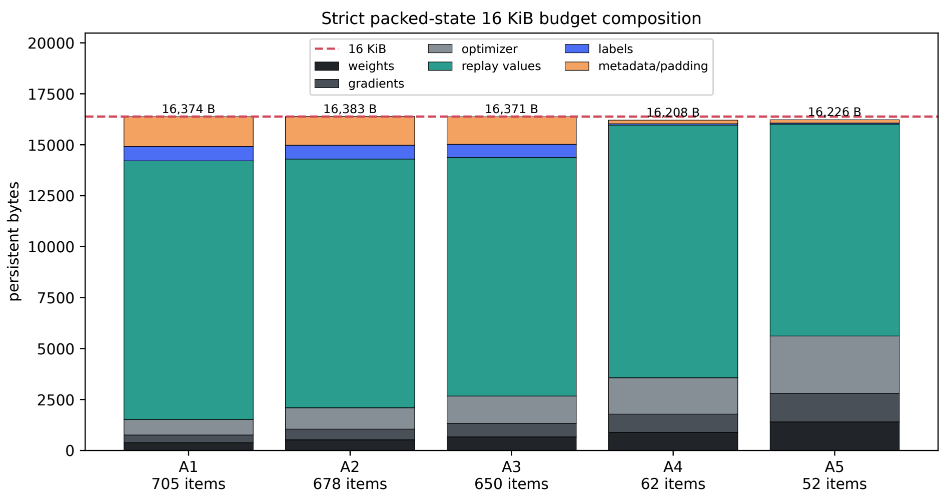

Figure 3: Persistent-state composition of each arm at 16 KiB. Volume arms (A1–A3) allocate almost the entire budget to replay values; depth arms (A4–A5) shift bytes into trainable weights, gradients, and optimizer moments, leaving room for only tens of raw exemplars.

4 Experimental Protocol

4.1 Source data and source training

The source backbone is trained *only* on the de Chazal DS1 partition of the MIT–BIH Arrhythmia Database. As published, DS1 and DS2 are *record*-disjoint rather than patient-disjoint (Section 4.2); the cohort correction below is what makes the evaluation set genuinely patient-disjoint from DS1. The model has 6,643 parameters and the architecture dimensions of Section 3. DS2 records never enter source training. On the held-out DS1 test split the frozen source attains macro-F1 = 0.749 over all five AAMI classes and 0.946 over the supported N/S/V classes, with strong V sensitivity but comparatively weak S behavior—the regime in which patient-specific adaptation is expected to help.

4.2 Primary and secondary cohorts

The de Chazal split is not patient-disjoint, and we correct it. The DS1/DS2 partition is the standard inter-patient protocol and we adopt it, but it contains one subject that appears on both sides. MIT–BIH record 202's own WFDB header states that it "was taken from the same analog tape as record 201" [39], and the two records carry identical demographics (68 M, identical medication list). Record **201 is in DS1** and record **202 is in DS2**. A source model trained on DS1 has therefore already seen this subject, and evaluating adaptation on record 202 as though it were an unseen patient measures partly memorization rather than personalization.

We audited all 48 MIT–BIH records for this failure mode, comparing header demographics and explicit same-tape annotations, and found exactly one such cross-split pair: 201/202. **We exclude record 202 from the evaluation cohort**, leaving **21 genuinely patient-disjoint records**. The source model is unchanged, since 201 remains a legitimate DS1 training record; only the evaluation set changes. Every primary and corrected analysis uses the 21-record cohort. One explicitly labelled historical sensitivity result retains the earlier 22-record split and is not compared with the primary results (Section 5.5). The primary grid is 6 arms $\times$ 21 records $\times$ 5 seeds = 630 adaptation cells. We flag this because the contaminated 22-record version is what a straightforward reading of the de Chazal protocol produces, and results computed that way are not strictly patient-disjoint and may be optimistically biased — on our own data, removing record 202 raised every arm mean slightly but cost the A4–A3 comparison its statistical significance.

Primary cohort. The 21 patient-disjoint DS2 records after the *post hoc* subject-identity correction form the primary cohort. Records are *not* selected using test-segment arrhythmia burden. This removes the selection bias of the earlier exploratory panel and is the cohort behind every headline result. Table 4 lists the record IDs, seeds, class supports, and split rule.

Secondary cohort. The 12-record arrhythmia-informative panel (records whose test window contains ≥ 30 arrhythmic beats) is retained only as secondary evidence for class-specific behavior, and is clearly labeled as such wherever it appears.

Table 4: Dataset, split, and preprocessing manifest for every database used. MIT–BIH records are one per subject *after* removing record 202, which shares a subject with DS1 record 201 and would otherwise break patient disjointness (Section 4.2). INCART contains multiple recordings per subject, so its statistical unit is the subject; the full per-record inclusion and exclusion manifests are released with the code. **The eligibility rules are not symmetric, and the difference matters:** primary MIT–BIH inclusion is *outcome-blind* — every corrected DS2 record is used and test-segment arrhythmia burden plays no part — whereas the external subsets additionally require ≥ 30 S+V beats *in the test window*, so external eligibility uses future test labels and enriches those cohorts for recordings where minority-class adaptation can be evaluated at all.

Field	MIT–BIH (primary)	INCART	SVDB
Role	source (DS1) + target (DS2)	external target	external target
Records evaluated	**21** (DS2)	11	46
Unique subjects	**21**	**6**	46
Records available	48	75	78
Lead	MLII	II	ECG1
Native sampling rate	360 Hz	257 Hz	128 Hz
Resampling	none	`resample_poly` to 360 Hz	
Beat window	198 samples, R-peak centred (99 left / 99 right)		
R-peak source	WFDB `atr` annotations		
Filtering	baseline-wander removal + low-pass		
RR normalization	MIT–BIH DS1 statistics only, clipped at ± 2		
AAMI mapping	N / S / V / F / Q; non-beat symbols ignored		
Adaptation protocol	chronological first 500 labels, gap_beats = 1, rest test		
Eligibility rule	**all corrected DS2; no test-burden filter**	≥ 30 S+V beats in the test window	≥ 30 S+V beats in the test window
Eligibility uses test labels?	**no** (outcome-blind)	**yes**	**yes**
Seeds	$\{42, 43, 44, 45, 46\}$		
DS1 records	101 106 108 109 112 114 115 116 118 119 122 124 201 203 205 207 208 209 215 220 223 230		
DS2 records	100 103 105 111 113 117 121 123 200 210 212 213 214 219 221 222 228 231 232 233 234 (*202 excluded: same subject as DS1 record 201*)		

4.3 Chronological adaptation

Each record is split in annotation (temporal) order. The chronologically first labeled beats form the adaptation set; a single unlabeled guard beat (gap_beats = 1) is dropped to prevent centered-window overlap between adaptation and test; all later beats are test-only. Early stopping uses only a slice of the adaptation union and never the test segment. Streamed patient beats are not counted as persistent replay storage—only the fixed source replay memory $\mathcal{R}_0$ is. Every patient resets to θ_0. The primary run therefore contains 6 arms $\times$ 21 records $\times$ 5 seeds = 630 adaptation cells (seeds $\{42, 43, 44, 45, 46\}$).

4.4 Signal preprocessing, stated exactly

Every record passes through the same front end before beat extraction, applied to the whole signal rather than per window. The pipeline is inherited unchanged from the source architecture we adopt [1], so that our adaptation results are attributable to the adaptation method rather than to a different front end. **Baseline-wander removal** uses two cascaded median filters: a 200 ms kernel strips the QRS and P complexes leaving baseline plus T-wave, a 600 ms kernel on that result strips the T-wave, and the remaining baseline estimate is subtracted from the original signal. Kernel lengths are rounded to the nearest odd sample count at the record's own sampling rate. **Noise suppression** is a 4th-order Butterworth low-pass at 35 Hz applied *zero-phase* (forward–backward, `filtfilt`), which is non-causal: it uses future samples and could not run unmodified in a streaming device. We keep it because it matches the source model's published front end, and we flag it as a deployment gap rather than a result. **Resampling** to 360 Hz uses polyphase resampling (`resample_poly`) and is applied to external databases only (INCART 257 Hz, SVDB 128 Hz); MIT–BIH is already at 360 Hz.

Non-causal preprocessing crosses the chronological boundary. All three of those operations — both median kernels and the zero-phase low-pass — are applied to the *whole record* before the chronological split, and all three are non-causal. An adaptation-side sample within the filters' backward reach of the split therefore depends on raw samples that belong to the test segment. Test *labels* are never involved, and no test beat enters adaptation, but the test *signal* is not strictly unseen, so the released protocol is properly described as **offline chronological label partitioning with non-causal whole-record preprocessing** rather than as strictly causal chronological adaptation.

We bounded the reach empirically rather than assuming it: injecting a truncation at a boundary and measuring how far the difference propagates backwards gives a one-sided influence of 84 samples at a 10^{-3} relative tolerance and 120 samples (333 ms) at 10^{-5} — shorter than a single RR interval. Because a beat window spans 198 samples, only the last one or two adaptation beats of a record can be affected at all. Section 5.7 reports a split-first rerun that removes the dependency entirely and quantifies what it changes. **Beat windows** are 198 samples centred on the annotated R peak, 99 left and 99 right; windows that would extend past a record boundary are dropped rather than padded. **R peaks** come from the WFDB `atr` annotations, never from a detector, so detector error is excluded from this study by construction. **Labels** follow the AAMI N/S/V/F/Q mapping; non-beat annotation symbols and symbols outside the mapping are skipped. **RR features** are the preceding and following intervals in seconds, normalized with MIT–BIH DS1 statistics only ($\mu = 0.7785\,\mathrm{s}$, $\sigma = 0.4956\,\mathrm{s}$) and clipped to ± 2; no target-database statistic ever enters normalization.

4.5 Replay quantization, stated exactly

Replay values are stored INT8 with **symmetric, per-tensor** quantization. For a stored tensor x the buffer holds

$$q = \mathrm{clip}(\mathrm{round}(x/s)\,, -127, 127)\,, \qquad \hat{x} = s\,q, \qquad s = \frac{\max|x|}{127}, \tag{7}$$

where the scale s is calibrated by absolute maximum over the whole buffer at the moment it is written, and the zero point is fixed at 0 by symmetry. One FP32 scale is stored per quantized tensor: one for a post-pool buffer, and two for a raw buffer, which quantizes the beat and the RR pair on separate scales. Per-row scales are deliberately *not* used, because a scale per exemplar would add 4 B per entry and is already charged as buffer metadata in Eq. (5). During adaptation the buffer is dequantized back to FP32 by Eq. (7) before entering the graph, so training sees exactly the INT8-rounded values a device would replay — the quantization error is inside the experiment, not idealized away.

4.6 Replay-buffer construction

The replay policy is held fixed across every arm so that the comparison isolates allocation rather than selection quality. The buffer is drawn once from the DS1 source pool by a **maximum-volume class-balanced selector**: each of the five AAMI classes receives an equal quota

$\lfloor N/5 \rfloor$, and because rare classes exhaust their pool before the quota is met, the unfilled slots are redistributed round-robin among classes that still have examples rather than being wasted. The selector is seeded by the run seed, so the buffer is *fixed across patients* within a seed and *resampled across seeds*; patient-to-patient variation therefore cannot come from a changing buffer, while seed-to-seed variation includes buffer variation, which is the conservative choice.

Values are stored INT8 with a per-tensor scale and zero point. A raw entry stores the ECG window *and* its RR pair, since the model consumes both and a window alone cannot be replayed; the honest per-item cost is therefore 200 B of values rather than the 198 B of the window alone, and the budget assertion uses 200. Post-pool entries are extracted once from the frozen source encoder and are therefore *stale by construction*: they are never re-extracted as adaptation proceeds, which is exactly why a post-pool buffer cannot track a representation that is itself changing — and is one reason a post-pool buffer pairs naturally with a frozen encoder. Raw entries carry no such staleness because they are re-forwarded through the current model on every update, at a greater but unmeasured compute cost (Appendix B). Replay and target beats are concatenated into one training set rather than mixed at a fixed per-batch ratio, so the effective replay ratio is $N_{\text{replay}}/(N_{\text{replay}} + N_{\text{target}})$. There is no eviction policy: the buffer is written once and never replaced, since the source distribution is static.

4.7 Metrics and statistical analysis plan

The primary metric, stated exactly. For patient p let $n_{p,c}$ be the number of test beats of AAMI class c. The scored class set and the primary metric are

$$\mathcal{C}_p = \{c \in \{\mathrm{N}, \mathrm{S}, \mathrm{V}\} : n_{p,c} \geq n_{\min}\}, \qquad F1_{\text{macro},p} = \frac{1}{|\mathcal{C}_p|} \sum_{c \in \mathcal{C}_p} F1_{p,c}, \tag{8}$$

with $n_{\min} = 1$: a class counts as present if the patient's test segment contains at least one beat of it. The remaining conventions are fixed as follows. The class set is N/S/V. Fusion (F) is excluded from the primary metric because it is absent from most DS2 test segments, so including it would make the denominator $|\mathcal{C}_p|$ vary for reasons unrelated to performance; F is reported separately as a pooled secondary metric, where the encoder arms show their largest relative gain. The unclassifiable class Q is excluded because the source model is not trained to separate paced and unclassifiable beats and the protocol makes no claim about them. A class with $n_{p,c} = 0$ is omitted from the average rather than scored as zero, since scoring an absent class as zero would penalize a patient for the composition of their own record. Precision and recall use the zero-division convention $0/0 = 0$, which applies only when a present class receives no predictions.

Secondary metrics are pooled accuracy, per-class S/V/F sensitivity, precision, and F1, the fraction of patients improved versus A0, and source-domain macro-F1 change Δ^{src} (Eq. 1), which is negative when adaptation costs source performance. Pooled and per-patient summaries are both reported because a single record (232) contributes a large fraction of the pooled S beats, so pooled scores alone would be misleading. Because the improved-patient count is sensitive to how small changes are treated, we report it both threshold-free and at a meaningful-change threshold of 0.02 macro-F1 (Table 5).

The primary metric, seed count, paired tests, effect-size definition, and Holm comparison family were fixed before the final runs. Record 202 was excluded after a subject-identity audit showed that it belongs to the same patient as DS1 record 201 (Section 4.2); all affected analyses were rerun without changing the methods or the comparison family.

Seeds are averaged per patient before patient-wise paired tests. For each planned comparison we report the mean and median paired difference, a **patient bootstrap** 95% confidence interval (10,000 resamples), a paired Wilcoxon signed-rank p-value, rank-biserial effect size, and a Holm correction across the six-comparison family {A4–A1, A5–A1, A4–A2, A4–A3, A4–A0, A5–A0}. A leakage and reproducibility audit (no DS1/DS2 record overlap, no adaptation/test window crossing, no *primary-cohort* test label used for patient inclusion, replay selection, early stopping, hyperparameter selection, or adaptation — external inclusion and the class-aware oracle deliberately do use test-window labels and are excluded from this check (Sections 5.9 and 5.10) — source-only normalization statistics, stored seeds and library versions, recorded source checkpoint hash) accompanies the run and returns PASS 11 / FIXED 1 / ASSESSED 1 / WARN 1 / FAIL 0.

The bootstrap, stated exactly. Every interval in this paper is a *patient bootstrap*, and we name it that way consistently because seeds are collapsed before resampling: (i) average the five seeds within each patient to give one score per patient; (ii) resample the 21 patients with replacement; (iii) recompute the arm mean or the paired difference on the resampled patients; (iv) repeat 10,000 times and take the 2.5th and 97.5th percentiles. Seeds are therefore not resampled, and we **do not** call this a hierarchical bootstrap — that term is reserved for a procedure that resamples seeds inside every resampled patient, which would be the right choice if seed variance were the quantity of interest. External-database intervals use the same procedure with subjects in place of patients.

5 Results

We organize the evidence around seven research questions. **RQ1:** where is replay memory-compressing? **RQ2:** does encoder plasticity improve patient adaptation? **RQ3:** is the gain caused by replay, plasticity, or both? **RQ4:** is the 16 KiB operating point itself a limitation? **RQ5:** does the result survive a class-reweighted source model? **RQ6:** does it transfer across databases? **RQ7:** how many labels are required?

Which results are pre-specified. RQ1–RQ3 and the source-domain change analysis follow a pre-specified plan: the primary metric (per-patient macro-F1 over present N/S/V classes), the unit of analysis (patient), the all-DS2 cohort, the seed count, the paired test, the effect size, and the six-comparison Holm family were all fixed in a written analysis plan before the final runs, and the six comparisons reported in Table 8 are exactly that pre-specified family with no additions. **One departure from a strict reading of "pre-specified" must be stated:** record 202 was excluded *after* the subject-identity audit described in Section 4, and every affected analysis was rerun. The exclusion changed the cohort, not the methods — no metric, test, effect size, or comparison family was altered in response to seeing results. RQ4–RQ7 are **secondary and exploratory**: they were specified after the primary analysis, they use reduced panels or different cohorts, and their p-values are not part of the corrected family. We flag the cohort and seed count at the head of each secondary subsection because they differ.

5.1 RQ1 — The replay-point cliff

Takeaway: only the post-pool representation is storage-compressive; every intermediate tap is several times larger than the raw beat, so the architecture fixes *where* replay can be cheap.

The cliff is a deterministic property of the architecture (Table 2, Fig. 2). A raw beat costs 198 B, the convolution/encoder tokens 1,056 B (5.3×), the FFN hidden tensor 8,448 B (42.7×), and only the post-pool+RR vector is smaller at 18 B (0.09×). Consequently, the frozen source model can afford 705 post-pool exemplars but only 62 raw exemplars in the same 16 KiB (Table 3). RQ1 confirms that the cliff constrains *where* replay is cheap; the remaining questions ask whether cheap replay is the best use of the budget.

5.2 RQ2 — Patient-level adaptation gain

Takeaway: both encoder-LoRA arms beat the frozen model with a large effect (Holm $p \leq 0.008$) and improve 17 of 21 patients, against 12 of 21 for maximum head-only replay, with stronger pooled minority-class performance.

Table 6 reports the all-DS2 primary result: per-patient macro-F1 (seeds averaged), the number of patients improved versus A0, and pooled S-class F1. Both encoder-LoRA arms dominate the frozen model and improve the most patients. A5 reaches the best per-patient mean (0.809 ± 0.179) and A4 the same 17/21 improved count with a slightly lower mean (0.803 ± 0.170); by contrast, maximum-volume head-only replay (A1) improves only 12/21 patients despite storing $\approx 11\times$ more exemplars. Fig. 4 shows the per-patient distribution and Fig. 5 the sorted per-patient gain; both make clear that the mean is carried by a minority of patients.

Patient safety, and what the improvement counts do and do not mean. Aggregate means hide the fact that adaptation is not uniformly beneficial, so we report the harm side explicitly, and we report it at two thresholds because the choice of threshold changes the story materially.

Counting any decline as harm, the encoder arms harm 4 of 21 patients and maximum head-only replay harms 8, which makes the encoder arms look safer. That impression does not survive a meaningful threshold. Defining the harm rate $R_{\text{harm}} = |\{p : G_p < -\epsilon\}|/P$ with $\epsilon = 0.02$ stated before analysis, the counts become A1 0/21 ($R_{\text{harm}} = 0.000$), A2 1/21, A3 2/21, A4 2/21, and A5 2/21 (Table 5) — so on this measure **maximum head-only replay is the *safest* arm**, harming nobody measurably, while the encoder arms each harm two patients. **Most of the apparent harm advantage of the encoder arms is sub-threshold noise**: at $\epsilon = 0.02$ the adapted arms harm zero (A1) to two (A3, A4, A5) patients each. The same correction applies to the improvement counts: the headline 17/21 for A4 and A5 falls to 12/21 when small changes are excluded, against 11/21 for A1. The encoder arms remain ahead on the improvement count, but by one patient rather than five, and they are *worse* on the harm count. What separates them is the size of the tail loss rather than its frequency: A4's worst patient loses 0.034 where A3's loses 0.052 and A5's 0.123.

We keep the threshold-free counts in Table 6 because they are what the pre-specified plan defined, and report the thresholded counts beside them so that neither is read alone. Either way the practical conclusion is the same: personalization helps most patients, measurably harms a few, and a deployment would need a per-patient acceptance check — adapt, evaluate on held-out adaptation-side data, and fall back to the frozen model if the gain is not positive — rather than an assumption that adaptation always helps.

Table 5: Patient-harm profile on the 21-record patient-disjoint cohort, seeds averaged. G_p is the per-patient change in the primary metric against the frozen model A0. Improved / unchanged / harmed use a meaningful-change threshold $\epsilon = 0.02$ stated before analysis, and $R_{\text{harm}} = |\{p : G_p < -\epsilon\}|/P$. "Impr. ($> 0$)" repeats the threshold-free count from Table 6.

Arm	Mean G_p	Med. G_p	Impr.	Unch.	Harmed	R_{harm}	Min G_p	5th pct.	Max G_p	Impr. (> 0)
A1	+0.119	+0.028	11	10	**0**	**0.000**	−0.017	−0.016	+0.793	12
A2	+0.124	+0.040	12	8	1	0.048	−0.069	−0.020	+0.789	15
A3	+0.109	+0.037	11	8	2	0.095	−0.052	−0.036	+0.770	13
A4	+0.136	+0.035	**12**	7	2	0.095	**−0.034**	−0.027	+0.804	**17**
A5	+0.143	+0.037	**12**	7	2	0.095	−0.123	−0.024	+0.804	**17**

Table 6: Per-patient macro-F1 over the 21-record patient-disjoint cohort, five seeds averaged within patient. Intervals are patient bootstraps (10,000 resamples).

Arm	Mean	SD	Median	95% CI
A0 (frozen)	0.666	0.245	0.666	[0.563, 0.766]
A1 (head + max replay)	0.786	0.177	0.747	[0.713, 0.860]
A2 (post-pool r=1)	0.790	0.176	0.766	[0.717, 0.863]
A3 (post-pool r=2)	0.775	0.169	0.710	[0.704, 0.847]
A4 (LoRA r=1)	0.803	0.170	0.760	[0.732, 0.875]
A5 (LoRA r=2)	0.809	0.179	0.786	[0.734, 0.884]

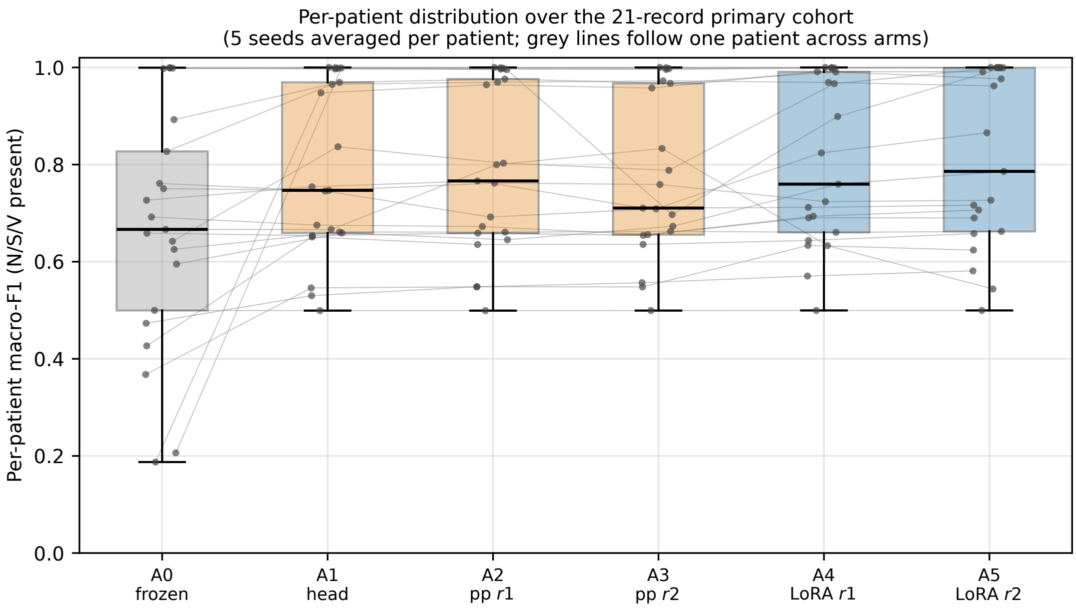

Figure 4: Per-patient macro-F1 across the 21-record primary cohort (five seeds averaged within patient, so each point is one patient). Boxes show the median and interquartile range; grey lines follow a single patient across arms. The encoder-LoRA arms (A4, A5) shift the distribution upward, but the spread is wide and heavily patient-dependent: the mean gain is not evenly distributed.

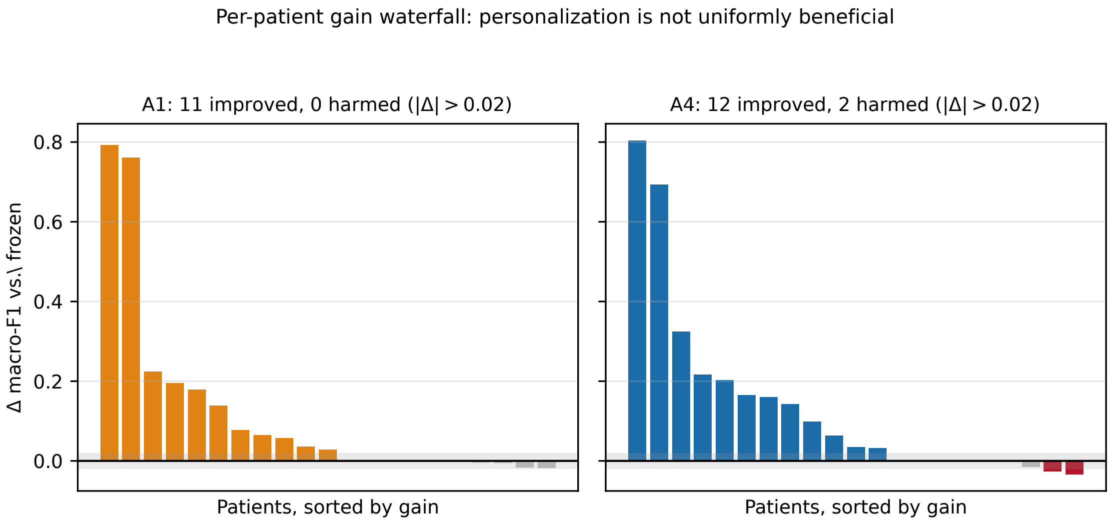

Figure 5: Per-patient gain against the frozen model, sorted, for maximum head-only replay (A1) and rank-1 encoder LoRA (A4) on the primary cohort. The grey band is the meaningful-change threshold $|\Delta| \leq 0.02$ macro-F1; bars inside it are indistinguishable from no change, and red bars are harmed patients. A small number of patients contribute most of the mean gain. At $\epsilon = 0.02$ the encoder arm (A4) harms two patients while maximum head-only replay (A1) harms none.

Pooled beat-level behavior tells the same story on minority classes (Table 7, Fig. 6). The frozen model labels the overwhelming majority of pooled S beats as normal (S F1 = 0.145); every adaptation arm recovers S, but the encoder arms are strongest, reaching S F1 0.832–0.835 and lifting fusion (F) F1 from 0.000 to 0.578–0.604 where the post-pool arms stall near 0.39. Pooled accuracy rises from 0.873 (A0) to 0.974 (A5).

Table 7: Pooled beat-level performance over the 21-record primary cohort (32,768 N / 1,385 S / 2,564 V / 291 F test beats). Each record contributes its seed-mean confusion matrix, so the supports are test beats rather than seed-multiplied decisions. Best in each column in bold.

Arm	Pooled acc.	S F1	V F1	F F1
A0	0.873	0.145	0.728	0.000
A1	0.962	0.776	0.915	0.416
A2	0.960	0.786	0.904	0.395
A3	0.960	0.773	0.909	0.392
A4	0.973	0.832	0.934	0.578
A5	**0.974**	**0.835**	**0.938**	**0.604**

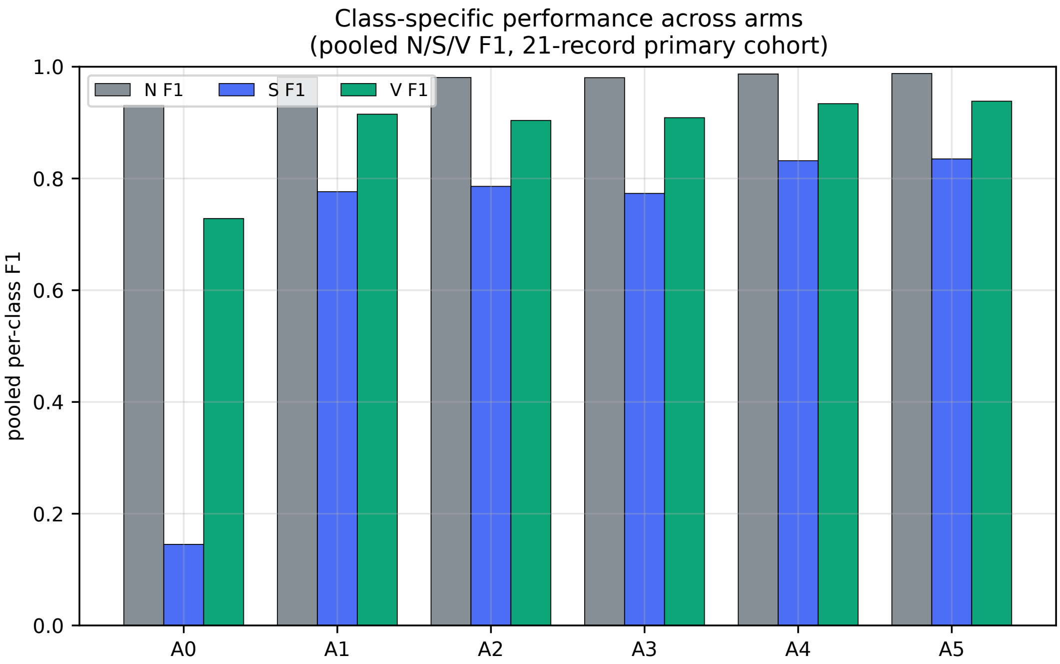

Figure 6: Class-specific pooled performance with the frozen baseline. Adaptation concentrates its benefit on the minority S class while preserving N and V; encoder arms give the largest minority-class gains. The fusion (F) class is not shown here because it is absent from most patients' test segments; pooled F1 for F is reported in Table 7, where the encoder arms show their largest relative improvement ($0.000 \rightarrow 0.58$–0.60 against ≈ 0.39 for the post-pool arms).

Direct paired tests. Table 8 and Fig. 7 report the six planned comparisons with Holm correction. Both encoder arms significantly beat the frozen model: A4–A0 mean +0.136 (95% CI $[0.055, 0.239]$, Holm $p = 0.007$, rank-biserial 0.77) and A5–A0 mean +0.143 (CI $[0.054, 0.252]$, Holm $p = 0.008$, rank-biserial 0.75). **Those are the only two comparisons that survive Holm correction.** The *direct* comparisons against maximum head-only replay are not significant: A4–A1 mean +0.017 (Holm $p = 0.25$) and A5–A1 mean +0.023 (Holm $p = 0.11$), with intervals including zero. A4 versus the rank-2 post-pool adapter A3 is also not significant after correction (Holm $p = 0.075$); on the contaminated 22-record cohort that contrast did clear the threshold ($p = 0.047$), and losing it is a direct consequence of removing the non-disjoint record. The honest reading is that encoder plasticity reaches significance against the frozen baseline *where post-pool volume does not*, and improves more patients, but direct superiority over maximum replay is **not** statistically established. This supports a replay–plasticity *allocation* frontier rather than "depth beats volume."

Table 8: The six pre-specified paired comparisons ($n = 21$). Bold marks the comparisons surviving Holm correction at $\alpha = 0.05$. **Only the two against the frozen model survive.**

Comparison	Mean Δ	Med. Δ	95% CI	Wilcoxon p	Holm p	r_{rb}	Impr./Wors.
A4 vs. A1	+0.017	+0.003	[−0.005, +0.039]	0.1259	0.2517	0.45	14 / 6
A5 vs. A1	+0.023	+0.003	[−0.002, +0.049]	0.0366	0.1097	0.58	16 / 4
A4 vs. A2	+0.013	+0.003	[−0.017, +0.043]	0.1790	0.2517	0.40	14 / 6
A4 vs. A3	+0.028	+0.008	[−0.006, +0.060]	0.0187	0.0747	0.64	16 / 4
A4 vs. A0	+0.136	+0.035	[+0.055, +0.239]	0.0012	0.0071	0.77	17 / 4
A5 vs. A0	+0.143	+0.037	[+0.054, +0.252]	0.0016	0.0080	0.75	17 / 4

Improved/worsened counts are threshold-free, matching the pre-specified plan; Table 5 repeats them at a meaningful-change threshold of 0.02. The A4–A1 and A5–A1 intervals include zero, which is the basis for this paper's refusal to claim superiority over maximum-volume replay.

Why an interval can span zero while the test is significant. Two rows pair a bootstrap interval containing zero with a raw Wilcoxon p below 0.05. This is not an inconsistency: the procedures target *different estimands*. The bootstrap interval estimates the *mean* paired difference, which a few large negative patients can drag toward zero, whereas the signed-rank test evaluates the distribution of signed *ranks* and is sensitive to a consistent direction of change even when magnitudes are small. Neither row is claimed as a positive result, since neither survives Holm correction.

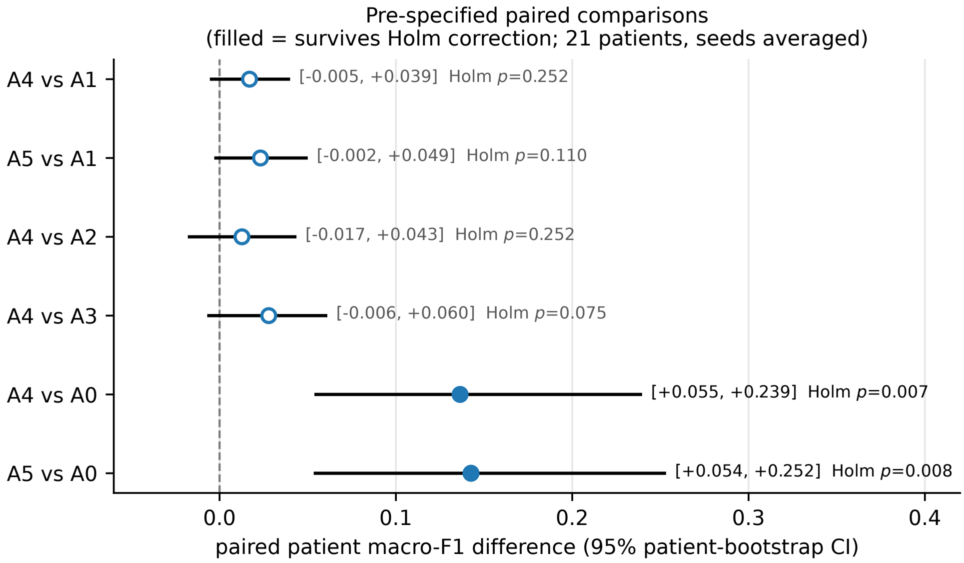

Figure 7: Forest plot of the planned paired comparisons (mean paired macro-F1 difference, patient bootstrap 95% CI). Encoder arms clear zero against the frozen model (A4–A0, A5–A0) but not against maximum head-only replay (A4–A1, A5–A1).

5.3 RQ3 — Controlled replay-versus-plasticity ablations

Takeaway: the encoder-plasticity contrast is statistically supported (B4−B2, $p = 0.005$, 16 of 21 patients improved), while replay *volume* is not; replay's clearest measured role is source retention, not target gain.

To separate replay location, replay volume, and trainable depth, we run the B0–B7 controlled ablation matrix (Table 10, Fig. 8). Arm means suggest a tidy story: adding raw replay to encoder LoRA lifts mean macro-F1 from 0.755 (B3, LoRA, no replay) to 0.803 (B4, LoRA + 62 raw), a +0.048 apparent contribution of *replay beyond plasticity*, while moving from a frozen-encoder head at the same raw-replay budget (B2) to encoder LoRA (B4) adds +0.030 from *plasticity beyond replay*.

Table 10: E9 controlled replay-versus-plasticity ablation matrix, recomputed on the 21-record patient-disjoint cohort (five seeds). **Arm means only** — the paired uncertainty that decides which contrasts are real is in Table 9, and the mean differences here must not be read as effects.

ID	Status	Scope	Replay	Items	Mean	vs B1
B0	complete	Frozen	none	0	0.666	−0.119
B1	complete	Head	postpool	705	0.786	0.000
B2	complete	Head	raw	72	0.773	−0.013
B3	complete	Encoder LoRA rank 1	none	0	0.755	−0.031
B4	complete	Encoder LoRA rank 1	raw	62	0.803	+0.017
B5	infeasible	Encoder LoRA rank 1	raw	72		
B6	complete	Head	postpool	62	0.778	−0.008
B7	complete	Post-pool adapter rank 1	postpool	678	0.790	+0.004

Only one of those two contributions survives paired testing. Differences of arm means are not evidence of a per-patient effect, so we test each contrast patient-wise (Table 9). The plasticity contrast is solid: B4−B2 gives +0.030 with a 95% interval of $[+0.007, +0.063]$ excluding zero, Wilcoxon $p = 0.005$, rank-biserial 0.74, and 16 of 21 patients improved against 4 worsened. **The replay contrast is not.** B4−B3 has a mean of +0.047 but a *median of exactly zero*, an interval $[-0.006, +0.112]$ that includes zero, $p = 0.473$, and a 10/11 split between patients helped and harmed — marginally more patients are *worse* with replay added than better. Its mean is carried by a handful of patients with large gains, not by a consistent effect.

The two remaining contrasts sharpen the paper's central point. Replay *volume* does almost nothing: B1−B6 compares 705 post-pool exemplars against 62 and yields +0.008 ($p = 0.232$) with only 5 patients improved and **15 worsened** — spending an extra 643 exemplars makes three times as many patients worse as better. Encoder depth at matched post-pool volume (B4−B7) is +0.013 and also not significant ($p = 0.179$).

We therefore state the controlled comparison result as: *controlled ablations show a statistically supported contribution from encoder plasticity, and a positive but statistically unsupported mean contribution from replay, with no measurable benefit from replay volume.* The earlier framing that replay and plasticity "contribute independently" overstated what the ablations establish, and we do not use it. Replay's clearest measured role in this paper is not target gain at all — it is source retention, where its effect is large and consistent (Section 5.4 and Section 5.11).

Table 9: Paired uncertainty for the four controlled contrasts on the 21-record patient-disjoint cohort, seeds averaged within patient. Arm means alone are misleading here: only the *plasticity* contrast is statistically supported. "Mean"/"Med." are the paired difference, "95% CI" a patient bootstrap interval over the 21 patients (10,000 resamples), p the Wilcoxon signed-rank value, r_{rb} the rank-biserial effect size, "I/W" the threshold-free improved/worsened split, and "@ϵ" the same split at the 0.02 threshold. B1, B4, and B7 are configuration-identical to A1, A4, and A2.

Contrast	Interpretation	Mean	Med.	95% CI	p	r_{rb}	I/W	@ϵ
B4−B3	replay beyond plast.	+0.048	−0.000	$[-0.006, +0.112]$	0.473	0.15	10/11	6/4
B4−B2	**plast. beyond replay**	+0.030	+0.006	$[\mathbf{+0.007}, \mathbf{+0.063}]$	**0.005**	**0.74**	**16/4**	6/0
B1−B6	replay volume	+0.008	−0.002	$[-0.027, +0.056]$	0.232	0.37	5/15	4/6
B4−B7	encoder depth	+0.013	+0.003	$[-0.019, +0.043]$	0.179	0.40	14/6	8/4

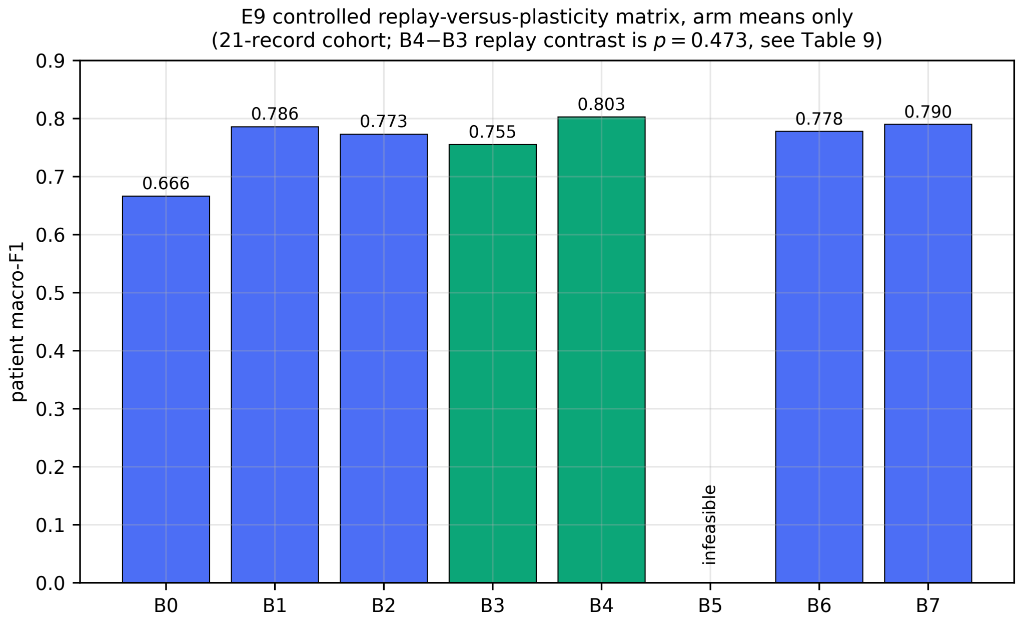

Figure 8: Replay-versus-plasticity ablation, arm means only. The apparent ordering B4>B3 (replay added) and B4>B2 (plasticity added) is what mean differences suggest, but **only the plasticity contrast survives paired testing**: B4−B2 has $p = 0.005$ while B4−B3 has $p = 0.473$ with a median of essentially zero and a 10/11 patient split. Read this figure together with Table 9, which carries the uncertainty; the bars alone overstate the replay effect.

5.4 Can regularization substitute for replay?

Takeaway: replay provides substantially better source retention than the tested zero-byte B-factor anchor under the evaluated configuration — at matched bytes the anchor loses about five times as much source performance as replay while adapting less. One regularizer was tested, so this is not a statement about regularization in general.

What this does and does not say about EWC-LoRA. The closest published method is EWC-LoRA [40], which applies a Fisher-based penalty to a shared low-rank update for task-sequential continual learning. **We do not implement it**, so nothing here should be read as a comparison against it. Two differences make a direct transfer non-obvious rather than merely untested: their setting accumulates *tasks* while ours is constrained by *bytes*, and a Fisher importance estimate is persistent state that would itself have to be charged against the 16 KiB ceiling — a diagonal Fisher over even the 223 trainable parameters of A4 is small, but over the full 6,643-parameter model it is not (Section 2). Evaluating EWC-LoRA under a byte budget is the single most relevant extension of the regularization arm of this study, and we flag it as such rather than claiming the question is closed.

The arms compared so far are all replay-based, which leaves open whether the budget would be better spent on a regularizer than on stored exemplars. The classical answer, elastic weight consolidation, is infeasible here: a full-model FP32 parameter anchor plus a Fisher value costs $6643 \times 8 = 53{,}144$ B, more than three times the entire budget before a single exemplar is stored.

A low-rank adapter admits a much cheaper anchor. The attention delta is $(xA)B$ with B zero-initialised, so shrinking B pulls the adapted model back toward the frozen source function, and it costs **zero persistent bytes** because the anchor *is* the frozen base weight the device already stores for inference. The implemented penalty is

$$\mathcal{L}_{\mathrm{reg}} = \lambda \sum_{m\in\{q,k,v,o\}} \|B_m\|_F^2, \qquad \lambda = 10^{-2}, \tag{9}$$

applied to the B factor of each of the four attention projections. We state one caveat plainly, because it bounds the interpretation: the quantity that actually determines the function is the product A_mB_m, and since $A_mB_m = (cA_m)(B_m/c)$, penalising B_m alone constrains the *parametrisation* rather than the function. $\|A_mB_m\|_F^2$ is the better-posed penalty; Eq. (9) is what was run, and we report it as such rather than describing it as a direct penalty on the source-function deviation. This is therefore **a low-cost memory-matched regularization baseline**, not the strongest regularizer the budget admits — restricted EWC on the adapter parameters, source-logit distillation, and output-consistency regularization all remain untested. We add two arms: R1 (rank-1 LoRA, anchor, no replay), byte-comparable with B3, and R2 (rank-1 LoRA, anchor, 62 raw exemplars), byte-identical to A4.

The answer is clear in both directions (Table 11). *As a substitute for replay, the anchor is weak.* Without replay it trades a little target gain for a little stability relative to unregularized LoRA ($0.755 \to 0.744$ target, Δ^{src} $-0.159 \to -0.136$). Replay achieves far more on both axes at once: A4 reaches $+0.136$ target gain against R1's $+0.078$, and holds the source-domain change to -0.027 against R1's -0.136, so the anchor loses about five times as much source performance as replay does, while also adapting less. *As a complement to replay, the anchor is redundant.* R2 is statistically indistinguishable from A4 on target performance (0.803 versus 0.803) and improves Δ^{src} only from -0.027 to -0.023.

We take two things from this, both hedged by the caveat above. First, the paper's replay-centric framing is not an untested assumption: at this scale, stored source exemplars regulate stability substantially better than this parameter-space penalty does, which is what one would expect when the shift is a change of patient rather than a change of task. Second, the anchor is nearly free, so R2 remains a reasonable default for a deployment wanting the small extra retention margin at no byte cost. Because only one regularizer was tested, and in its scale-ambiguous form, we claim only that *this* anchor does not substitute for replay — not that regularization in general cannot.

Table 11: Memory-matched regularization baseline on the 21-record patient-disjoint cohort (5 seeds). The source anchor is $\lambda \sum_m \|B_m\|_F^2$ on the B factor of each attention projection (Eq. 9); because B is zero-initialised and the frozen base weights *are* the anchor, it costs **zero persistent bytes**, so R1 is byte-comparable with B3 and R2 with A4. Penalising B rather than the product AB is scale-ambiguous and constrains the parametrisation rather than the function; this is a low-cost baseline, not the strongest regularizer available. Δ^{src} is the signed source-domain macro-F1 change.

Arm	Mechanism	Replay	Anchor	Bytes	macro-F1	Δ vs. A0	Δ^{src}
A0	frozen	—	—	0	0.666	—	0.000
B3	LoRA r=1	—	—	3,568	0.755	+0.089 ($p = 0.044$)	−0.159
R1	LoRA r=1 + anchor	—	yes	3,568	0.744	+0.078 ($p = 0.052$)	−0.136
A4	LoRA r=1 + replay	62	—	16,208	0.803	+0.136 ($p = 0.0012$)	**−0.027**
R2	LoRA r=1 + both	62	yes	16,208	**0.803**	**+0.137** ($p = 0.0011$)	−0.023

5.5 RQ4 — Adam budget sweep for rank-1 and rank-2 adaptation

Takeaway: within the measured 12-record Adam rank-1/rank-2 panel, raising the adaptation-state budget from 8 to 64 KiB produced no statistically detectable benefit (secondary evidence, reduced panel). We do not generalize beyond the panel that was measured.

The primary study fixes a single 16 KiB budget, which raises the obvious question of whether the conclusions are an artifact of that particular point. The experiment that speaks to it, E10, was designed as a full budget × optimizer × rank grid but was executed only in part, so we name it for what was actually measured: an **Adam budget sweep for rank-1 and rank-2 adaptation** at 8, 16, 32, and 64 KiB. The unmeasured dimensions are not discussed as results anywhere in this paper.

Scope, stated up front. This sweep is **secondary evidence and is not comparable to the primary cohort**. Two limitations must be read before the numbers. First, to keep the grid tractable it was executed on a reduced panel of **12 DS2 records and 3 seeds** (864 adaptation cells), not the 21-record, 5-seed primary cohort; absolute values are therefore not interchangeable with Table 6, and no cell of this sweep may be ranked against a primary-cohort arm. Second, the executed grid is a *subset* of the planned design: 49 of 140 planned configurations are measured, 5 are analytically infeasible at their budget, and 86 remain unrun. The budget × scope × rank surface is complete *only for Adam at ranks 1 and 2*; SGD is measured at 8 KiB alone, SGD-with-momentum not at all, and ranks 4 and 8 are almost entirely unmeasured (rank 8 is infeasible at 8 and 16 KiB by the memory model).

The single claim this experiment supports is therefore narrow and we state it once, here: *within the measured Adam rank-1/rank-2 panel, no statistically resolvable improvement was observed when increasing the persistent-state budget from 8 to 64 KiB.* We make **no optimizer-ranking claim, no rank-4 or rank-8 claim, and no claim that the 140-cell design is complete.**

Result: no detectable benefit above 8 KiB. Within the Adam panel (Table 12, Fig. 9), raising the budget from 8 to 64 KiB does not improve per-patient macro-F1 for any family. The encoder-LoRA family is flat-to-declining (0.805 → 0.790 → 0.776 → 0.768), the post-pool-adapter family varies within 0.762–0.780, and the head-only family within 0.762–0.774.

Overlapping confidence intervals are not evidence of equivalence, so we test the budget effect directly. Holding optimizer, trainable scope, rank, and replay location fixed and varying *only* the budget gives a paired contrast on each record; we test each with a Wilcoxon signed-rank test and, for equivalence, with a two-one-sided-tests (TOST) procedure against a margin of $\delta = 0.05$ macro-F1 stated before analysis (Table 13). Of the 18 budget contrasts, **14 met the pre-specified ± 0.05 equivalence criterion** against their 8 KiB reference, and none of the raises is significantly positive.

Why $\delta = 0.05$, and what the 18 decisions are not. The margin was fixed before analysis at 0.05 macro-F1 because it is roughly a third of the primary adaptation effect this paper is built on (A4−A0 = +0.136): a budget change that moves macro-F1 by less than δ cannot account for the adaptation gain, which is the question the sweep exists to answer. It is a within-panel yardstick, not a claim about clinical meaningfulness, and it is coarser than the $\epsilon = 0.02$ per-patient harm threshold used in RQ2 because the sweep panel has 12 records rather than 21 and correspondingly wider intervals. **The 18 TOST decisions are not multiplicity-adjusted.** They sit outside the six-comparison Holm family, which was reserved for the primary analysis, so each equivalence decision is an uncorrected per-contrast statement and the count "14 of 18" should be read descriptively. Adjusting would make equivalence harder to declare, not easier, so the direction of the resulting bias is toward over-claiming equivalence — another reason we keep the claim inside the measured panel.

Two contrasts deserve explicit mention because they cut against a naive "more memory is at least harmless" reading. For rank-2 encoder LoRA, moving from 8 to 64 KiB *significantly reduces* macro-F1 by 0.039 (Wilcoxon $p = 0.003$), and the 32 KiB contrast is also negative (−0.027,

$p = 0.027$); neither is equivalent within the margin. The likely mechanism is that a raw-replay buffer of several hundred source exemplars starts to dominate a 500-beat target set, pulling the adapted model back toward the source. We report this as a measured effect on the secondary panel rather than a general law.

The defensible reading is a null result of a useful kind, stated at the scope it was measured: *within the measured 12-record Adam rank-1/rank-2 panel, increasing the adaptation-state budget produced no statistically detectable benefit, and for rank-2 encoder LoRA it produced a detectable cost.* On that panel, extra persistent memory did not buy accuracy, which is consistent with the allocation question studied in RQ2 and RQ3 — how to spend a fixed budget — being the one that matters. We do not extend this to unmeasured optimizers, ranks, or cohorts, and we do not claim that 16 KiB is provably sufficient in general.

A separate low-byte measurement. Independently of this sweep, a fixed-scope run on the *full* 22-record, 5-seed cohort measured a rank-1 encoder-LoRA configuration holding only **8 raw replay samples** in 5,246 B — under a third of the budget — reaching 0.801 per-patient macro-F1 on the contaminated 22-record cohort, against 0.802 for the 16 KiB rank-2 arm (A5) on the same cohort. We quote this pair on the 22-record basis because that fixed-scope run was not repeated after the cohort correction. **It is therefore a historical sensitivity result and is *not* directly comparable with Table 6**, which is computed on the corrected 21-record cohort; the two numbers quoted here are internally comparable with each other and with nothing else in this paper. Read only within itself, the pair is suggestive of the same saturation direction — a third-of-budget configuration landing beside a full-budget one — but it carries no weight against any primary-cohort figure.

Table 12: Adam budget sweep, rank-1 and rank-2 (E10): best configuration per family at each budget (per-patient macro-F1, N/S/V present classes). **Reduced secondary panel of 12 DS2 records and 3 seeds, Adam only, ranks 1 and 2 — not comparable to the primary cohort of Table 6.** SGD and SGD-with-momentum cells are excluded as unmeasured (except 8 KiB SGD). Formal budget contrasts and equivalence decisions are in Table 13; the means here are descriptive and should not be read as a ranking.

Family	8 KiB	16 KiB	32 KiB	64 KiB
Encoder LoRA + raw replay	**0.805**	**0.790**	0.776	**0.768**
Post-pool adapter + post-pool	0.762	0.765	**0.780**	0.767
Head only	0.762	0.774	0.765	0.763

Table 13: Paired budget contrasts within the Adam rank-1/rank-2 sweep (E10). Optimizer, trainable scope, rank, and replay location are held fixed and only the budget varies, so each row is a paired comparison over the same 12 records of the secondary panel (3 seeds, Adam only). Δ is the mean paired change in per-patient macro-F1 relative to the same configuration at 8 KiB. "Equiv." is a TOST equivalence decision at the pre-stated margin $\delta = 0.05$. Raising the budget is never significantly beneficial; for rank-2 encoder LoRA it is significantly harmful.

Scope	Replay	Rank	Budget	Δ vs. 8 KiB	Wilcoxon p	Equiv.
Encoder LoRA	raw	1	16 KiB	+0.015	0.204	yes
			32 KiB	+0.007	0.424	yes
			64 KiB	+0.002	0.380	yes
Encoder LoRA	raw	2	16 KiB	−0.013	0.164	yes
			32 KiB	−0.027	**0.027**	no
			64 KiB	−0.039	**0.003**	no
Head only	post-pool	0	16 KiB	+0.006	0.577	yes
			32 KiB	+0.004	0.898	yes
			64 KiB	+0.003	0.831	yes
Head only	raw	0	16 KiB	+0.000	0.519	yes
			32 KiB	−0.004	0.677	yes
			64 KiB	+0.010	0.776	no
Post-pool adapter	post-pool	1	16 KiB	+0.016	0.240	yes
			32 KiB	+0.030	0.520	no
			64 KiB	+0.017	0.831	no
Post-pool adapter	post-pool	2	16 KiB	−0.001	1.000	yes
			32 KiB	+0.007	0.520	yes
			64 KiB	+0.006	1.000	yes

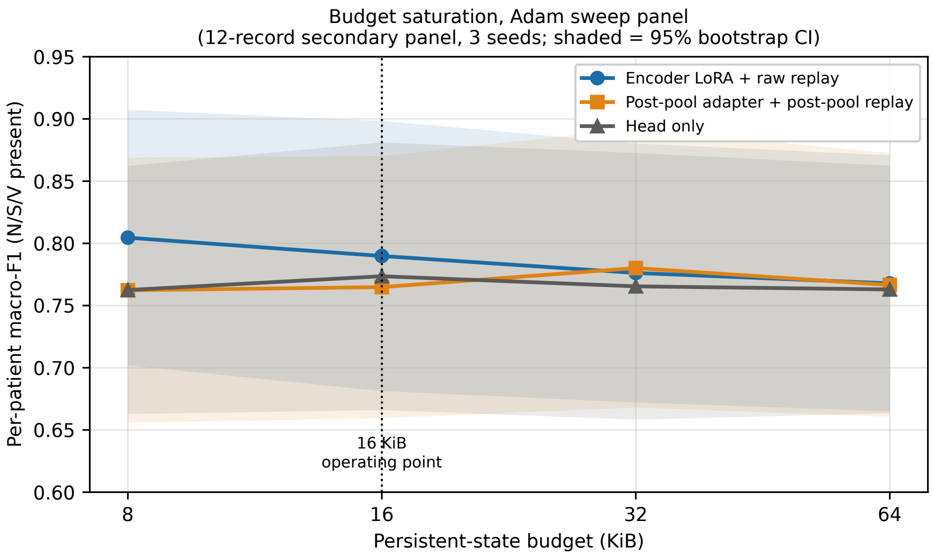

Figure 9: Per-patient macro-F1 versus persistent-state budget for the Adam rank-1/rank-2 sweep (E10). **Secondary cohort: 12 DS2 records, 3 seeds, Adam only, ranks 1 and 2; not directly comparable to Table 6.** Points are the best configuration per family at each budget and shaded bands are 95% bootstrap intervals over records. The bands overlap, but overlap alone is not evidence of equality — the supporting claim is the paired TOST analysis of Table 13, in which 14 of 18 budget contrasts are equivalent within ±0.05 macro-F1 and none is significantly positive.

5.6 Does the result depend on spending the last kilobyte?

Takeaway: the result does not depend on saturating the budget — reserving 1 KiB and regenerating replay counts leaves every adapted arm significantly above frozen ($p \leq 0.010$) with encoder adaptation still among the top allocations.

Every configuration reported so far fills the 16 KiB budget almost exactly — A2 reaches 16,383 of 16,384 bytes. That is not a deployable margin: real firmware needs room for a version field, a checksum, and future additions, and a configuration generated against a ceiling it exactly saturates would break the moment anything is added. We therefore re-ran the entire A0–A5 grid, same 21 records and five seeds, against an effective ceiling of $B_{\text{budget}} - B_{\text{reserve}} = 16{,}384 - 1{,}024 = 15{,}360\,\text{B}$, regenerating replay counts rather than reporting old results against new byte figures. The reserve costs A1 49 exemplars, A4 five, and A5 five (Table 3).

The qualitative conclusion is preserved (Table 14): every adapted arm remains significantly above the frozen model ($p \leq 0.010$), encoder adaptation remains among the top-performing allocations, and A4 is unchanged at 0.802. The *exact* arm ordering is not preserved, and we say so rather than claiming otherwise: without the reserve the order is A5 > A4 > A2 > A1 > A3, and with it A4 > A1 > A5 > A2 > A3. A1 gains slightly (0.786 → 0.793) while A5 loses more (0.809 → 0.792). A5 is thus more sensitive than A4 to losing the last few replay samples, which supports **rank-1 encoder LoRA as the more robust engineering operating point** — a practical conclusion the unreserved analysis alone would have missed. We keep the unreserved configuration as the primary analysis because it is what the pre-specified plan defined, and report the reserve replication as the deployable-margin check. **The primary statistical conclusion is unchanged, although the exact ordering and the preferred engineering operating point change under the reserve.**

Table 14: Implementation-reserve replication on the 21-record patient-disjoint cohort. The full A0–A5 grid re-run against an effective ceiling of 15,360 B, i.e. holding back a 1 KiB firmware reserve, same records and five seeds. Replay counts fall by roughly 7–8%. Every arm still significantly beats the frozen model and encoder adaptation stays among the top allocations, though the exact ordering changes: A5 loses more than A4, making rank-1 the more robust operating point.

	16,384 B ceiling			15,360 B ceiling (+1 KiB reserve)		
Arm	Items	macro-F1	Δ vs. A0	Items	macro-F1	Δ vs. A0
A0	0	0.666	—	0	0.666	—
A1	705	0.786	+0.119	656	0.793	+0.127
A2	678	0.790	+0.124	629	0.779	+0.113
A3	650	0.775	+0.109	601	0.771	+0.105
A4	62	0.803	+0.136	57	**0.802**	**+0.136**
A5	52	**0.809**	**+0.143**	47	0.792	+0.125

All reserve-arm gains over A0 remain significant by paired Wilcoxon ($p = 0.006$, 0.003, 0.010, 0.004, 0.005 for A1–A5). Unreserved the order is A5 > A4 > A2 > A1 > A3; reserved it is A4 > A1 > A5 > A2 > A3.

5.7 Does the boundary filter dependency matter?

Takeaway: no. Removing the dependency entirely moves every arm mean by at most 0.007 macro-F1, leaves both encoder-versus-frozen contrasts significant at $p = 0.0012$, and leaves both encoder-versus-A1 contrasts non-significant. The conclusions are not artifacts of non-causal preprocessing.

The released preprocessing filters each record as a whole before the chronological split (Section 4), so adaptation-side samples within the filters' backward reach of the boundary can depend on test-segment raw samples. To remove that dependency rather than argue about its size, we rebuilt the beat store *split-first*: the raw signal is cut at the boundary, a 720-sample (2 s) guard is discarded from each side — roughly a sixfold margin on the measured 120-sample influence reach — and the two pieces are filtered independently, so no adaptation-side sample can see a test-side raw sample. Beats whose window would straddle the cut are dropped, which costs one to three adaptation beats per record; we then truncate every record to a uniform 497-beat adaptation block so the protocol stays balanced.

We reran A0, A1, A4, and A5 on this store over the same 21 patient-disjoint records and the same five seeds. Table 15 reports the result.

Three things are worth separating. First, the **frozen arm is unmoved** (−0.0000, largest per-record shift 0.0004). A0 performs no adaptation, so it isolates the effect of refiltering the test segment alone, and that effect is numerically negligible — which is the expected result and a useful check that the rebuild did not perturb the evaluation data.

Second, the **adapted arms move very little, and not in the direction leakage would predict**. A1 rises by 0.007, A4 falls by 0.006, and A5 is unchanged to three decimals. If whole-record filtering had been inflating the results, removing it should have lowered them systematically; instead the shifts have mixed sign. **The seeds are identical across the two runs, so these are not run-to-run fluctuations.** They reflect the combined effects of the three things the rerun actually changes: independent segment filtering, removal of boundary-adjacent beats, and reducing the adaptation block from 500 to 497 labelled beats. Their magnitude is too small to alter any inferential conclusion.

Third, and most importantly, **no inferential conclusion changes**. Table 16 recomputes the four contrasts that matter on the split-first cohort, reporting both raw and family-corrected p-values. These four form their own sensitivity family and are *not* pooled with the six pre-specified primary comparisons. Both encoder arms remain significantly above the frozen model with identical improved-patient counts (17/21), and the two comparisons against maximum head-only replay remain non-significant, so the paper's central refusal to claim superiority over A1 is unaffected. The one substantive change is that the *margin* over A1 shrinks — A4−A1 falls from +0.017 to +0.004 and A5−A1 from +0.023 to +0.017 — which weakens the encoder-over-head ordering slightly rather than strengthening it. We report that in the direction it points: the split-first evidence makes the frontier reading *more* appropriate, not less.

This is a sensitivity analysis, not a replacement primary analysis. The headline numbers throughout the paper remain those of the released pipeline, so that they correspond to the pre-specified plan and to the published artifacts; the split-first rerun is reported here as the check that the boundary dependency does not carry the result. A fully causal streaming front end — which would also have to replace the zero-phase filter, not merely the record boundary — remains future work.

Table 15: Boundary-leakage sensitivity (E17). The split-first rerun cuts the raw signal at the boundary, discards a 2 s guard each side, and filters the pieces independently, so no adaptation-side sample can depend on a test-side raw sample. **Every conclusion is preserved.**

Arm	Whole-record	Split-first	Δ
A0 (frozen)	0.666	0.666	−0.000
A1 (head + max replay)	0.786	0.793	+0.007
A4 (LoRA r=1)	0.803	0.796	−0.006
A5 (LoRA r=2)	0.809	0.809	+0.000

Table 16: Paired contrasts recomputed on the split-first (E17) cohort, $n = 21$ patients, five seeds averaged within patient. **These four sensitivity contrasts form their own family and are not pooled with the six pre-specified primary comparisons**; the Holm column corrects within these four only. Both frozen-model contrasts remain significant and neither encoder arm clears maximum head-only replay, matching the primary cohort.

Contrast	Mean Δ	Med. Δ	95% CI	Raw p	Holm p	Impr./Wors.
A4−A0	+0.130	+0.040	[+0.050, +0.231]	0.0012	0.0047	17 / 4
A5−A0	+0.143	+0.039	[+0.060, +0.245]	0.0012	0.0047	17 / 4
A4−A1	+0.004	+0.001	[−0.021, +0.027]	0.3135	0.4359	14 / 6
A5−A1	+0.017	+0.001	[−0.014, +0.047]	0.2180	0.4359	14 / 6

"Raw p" is the unadjusted paired Wilcoxon signed-rank value; "Holm p" corrects across these four sensitivity contrasts. Intervals are patient bootstraps (10,000 resamples).

5.8 RQ5 — Robustness to alternative class-reweighted source checkpoints

Takeaway: the adaptation gain persists under two alternative class-reweighted source-training objectives. This supports robustness to the source *objective*, but does **not** establish robustness to a *stronger unseen-patient source model*, which was never demonstrated (secondary evidence).

A key robustness question is whether the adaptation gain merely corrects an unusually weak S-class source. We train five architecture-identical source variants on MIT–BIH DS1 with five seeds—cross-entropy, effective-number class weighting, and focal loss $\gamma \in \{1, 2, 3\}$—and measure them under the same protocol (Table 17).

We do not call these checkpoints stronger, because on the reported metric they are not. Intra-patient DS1 S sensitivity is 0.881 for plain cross-entropy, 0.843 for focal γ=2, and 0.776 for effective-number weighting: the class-reweighted objectives are *below* the cross-entropy baseline on exactly the axis that motivated the concern, and they also cost overall macro-F1. Calling them *stronger source checkpoints* would not be supported by these results, so we describe them for what they are — **alternative source checkpoints trained with class-reweighted objectives** — and treat RQ5 as a test of whether the adaptation result is *robust to the choice of source objective*, not as adaptation from a stronger starting point.

One further caution: the S sensitivities above are measured on an intra-patient DS1 split, whereas the frozen-source weakness that motivates adaptation is an unseen-patient DS2 property. These are different domains and we do not compare them as if they were the same. Establishing which checkpoint is genuinely stronger on unseen patients would require evaluating every variant frozen on DS2, which we have not done; that is the experiment RQ5 would need to make a true stronger-source claim.

Training five variants is not the same as adapting from five, and we separate the two explicitly: all five variants were *trained* and measured for source quality (Table 17), while adaptation was re-run from **two** of them — effective-number weighting and focal γ=2, the two that most change the minority-class behaviour the concern is about (Table 18). The gain survives: adapting from the effective-number source lifts per-patient macro-F1 (N/S/V) from 0.458 (frozen) to 0.599 (both head-only and encoder LoRA), and from the focal-γ2 source from 0.515 to 0.642 (head) and 0.644 (encoder LoRA). On the encoder-versus-head question these two runs say less than they appear to: *the mean encoder-minus-head direction remains nonnegative under two alternative class-reweighted source objectives, but the margins are small (*$+0.0001$ *and* $+0.0024$*) and no uncertainty is reported*, so they are consistent with the primary-cohort finding of a non-significant direct comparison rather than independent support for an ordering. Adaptation from the remaining three trained variants (cross-entropy and focal $\gamma \in \{1, 3\}$) is not reported here, so we claim only that *the ordering was re-evaluated from two class-reweighted source checkpoints*, not from five. The three RR-architecture and CNN variants require new model graphs and were never trained; they are labeled `not_implemented_not_run` in the ledger.

Table 17: E11 source-checkpoint ledger. All five variants share the architecture of the shipped backbone and differ only in the training objective; each is trained on MIT–BIH DS1 with five seeds and scored on a held-out *intra-patient* DS1 split. The right-hand column separates the checkpoints we *trained* from those we also *adapted from*, because the robustness claim covers only the latter. **None of the reweighted objectives beats plain cross-entropy on S sensitivity** (0.881), so we do not describe them as stronger sources; RQ5 tests robustness to the source objective, not adaptation from a better starting point. These intra-patient DS1 figures are also not comparable to unseen-patient DS2 behaviour.

Source variant	Status	macro-F1	S sens.	Adapted from?
Cross-entropy (`original_ce`)	measured, 5 seeds	**0.742**	**0.881**	no
Effective-number weighting	measured, 5 seeds	0.618	0.776	**yes**
Focal loss $\gamma = 1$	measured, 5 seeds	0.741	0.878	no
Focal loss $\gamma = 2$	measured, 5 seeds	0.730	0.843	**yes**
Focal loss $\gamma = 3$	measured, 5 seeds	0.725	0.780	no
RR-conditioned token embedding	not implemented	—	—	—
RR token before attention	not implemented	—	—	—
Lightweight inter-patient CNN	not implemented	—	—	—

The three unimplemented rows require new model graphs and were never trained; they are recorded here rather than omitted so the ledger matches the experiment plan. Three further pre-existing `beat_classifier` checkpoints exist in the repository but were measured on an intra-patient random split rather than the DS1-only protocol, so they are not comparable and are excluded from this table.

Table 18: Adaptation from two E11 class-reweighted source checkpoints (per-patient macro-F1 over N/S/V, DS2, five seeds). The adaptation gain persists under a changed source objective, which is the robustness question RQ5 answers. The $\Delta_{\mathrm{A4-A1}}$ column is nonnegative in both cases but the margins are small and **no uncertainty is reported for them**; it must not be read as evidence of an encoder-over-head ordering.

Source-training objective	Frozen	Head (A1)	Encoder LoRA (A4)	$\Delta_{\mathrm{A4-A1}}$
Effective-number weighting	0.458	0.599	0.599	+0.0001
Focal loss $\gamma = 2$	0.515	0.642	0.644	+0.0024

5.9 RQ6 — Cross-database validation

Takeaway: per-subject adaptation transfers across databases — INCART (6 subjects) and SVDB (46) both show positive gains, with encoder LoRA again with a higher mean than head-only replay on both external subsets (secondary evidence, small INCART cohort; no paired direct-comparison interval is reported).

External validity is evaluated on INCART (11 recordings from six unique subjects) and the MIT–BIH Supraventricular Arrhythmia Database (SVDB, 46 records), with a documented pre-processing manifest (lead, resampling, R-peak source, beat window, RR normalization, AAMI mapping, exclusions). The same chronological, target-informed protocol is applied per record. In contrast to a database-level aggregate evaluation—which is neutral-to-negative because it performs no per-record adaptation—the per-record result is clearly positive (Table 20, Fig. 10).

Statistical unit: unique subjects, not recordings. INCART contains multiple recordings per subject, so a record-level analysis would treat non-independent recordings as independent samples. We parsed the subject identifier from every INCART header and grouped accordingly: the 11 evaluated records collapse to **six unique subjects** (`I01--I02` are one 65-year-old subject, `I03--I05` a second, `I06--I07` a third, `I08` a fourth, `I09--I10` a fifth, and `I13` a sixth). Because the saved cross-database cells retain the source record name, we re-aggregated the existing runs to one score per subject—mean over that subject's recordings and seeds—and bootstrap over subjects rather than recordings. All INCART numbers below are subject-level. SVDB recordings are one per subject, so its unit is unchanged. Every configured recording in both databases now carries exactly one deterministic disposition (Table 19).

What is adapted, and what is aggregated. These two things are not the same and the distinction matters. Adaptation is **record-wise**: each recording starts from the same frozen source checkpoint, adapts on its own early chronological segment, and is tested on its own later segment, with the model *reset between recordings*. Only the resulting scores are then averaged within a subject, and the bootstrap resamples subjects. The correct description is therefore *record-wise chronological adaptation, aggregated to the subject level for statistical analysis*. We do **not** carry one persistent model across a subject's recordings, so nothing here should be read as evidence about a single long-lived patient model that accumulates across sessions. True subject-level adaptation—order a subject's recordings chronologically, adapt once on the earliest, test on the later ones—is a different and stronger experiment, and it is left to future work. Grouping to subjects fixes the *independence* defect in the statistics; it does not convert the protocol into a longitudinal one.

The external gain survives the correction. On INCART, subject-level macro-F1 (N/S/V) rises from 0.459 (frozen) to 0.595 with head-only replay (+0.137, 95% CI $[+0.058, +0.215]$), 0.618 with encoder LoRA and raw replay (+0.160, CI $[+0.082, +0.234]$), and 0.647 with no-replay encoder LoRA (+0.189, CI $[+0.110, +0.264]$); **6 of 6 subjects improve under every method** and no interval includes zero. Collapsing recordings to subjects lowers the encoder-LoRA gain from the record-level +0.194 to +0.160—the record-level figure over-weighted the multi-recording subjects—but does not change the conclusion. With $n = 6$ the inferential power is limited, so we treat the INCART result as descriptive evidence with an interval attached rather than as a significance claim.

On SVDB it rises from 0.384 to 0.580/0.610/0.626 (+0.196/+0.226/+0.242), with 45–46 of 46 subjects improved and intervals excluding zero (Table 20, Fig. 10). Two observations matter. First, the replay–plasticity ordering *reproduces externally and with a larger margin* than on DS2: encoder LoRA beats maximum head-only replay by +0.023 on INCART and +0.029 on SVDB. Second, dropping replay entirely yields the *largest* external target gain but also the most source-domain change (-0.11 on INCART, -0.14 on SVDB, versus ≤ -0.07 with replay), a textbook stability–plasticity trade-off: under strong domain shift, replay trades a little target gain for markedly better source retention.

One caveat is not statistical but selective. The INCART cohort is a tractability-capped convenience sample: the configuration attempts only the first 13 of 75 records, of which one fails preprocessing and one lacks minority-class support. The 62 records outside that cap were never attempted, so the INCART cohort is not a principled selection from the database and should not be read as representative of it. The SVDB cohort is capped at 54 of 78 recordings on the same basis, with 8 of the attempted 54 excluded for insufficient S+V support in the test window.

Table 19: External-database inclusion and exclusion flow. **External evaluation uses convenience-capped, minority-support-filtered subsets. The eligibility rule uses test-window class labels** — a recording is retained only if its later test window carries at least 30 S+V beats — **and therefore enriches the cohort for recordings where minority-class adaptation can be evaluated at all.** These are consequently not representative full-database validations and no claim of representative external validation is made from them. Every configured recording nevertheless carries exactly one deterministic disposition; the full per-record manifests (lead, beat counts, class support, and reason) are released with the code as `incart_inclusion_exclusion_manifest.csv` and `svdb_inclusion_exclusion_manifest.csv`.

Stage	INCART		SVDB	
	Records	Subjects	Records	Subjects
Configured in the database	75	32	78	78
Attempted (within record cap)	13	7	54	54
Preprocessing successful	12	6	54	54
Meets minority-class requirement	11	6	46	46
Final evaluated cohort	**11**	**6**	**46**	**46**
Exclusion reasons				
Outside configured record cap	62	—	24	24
Preprocessing failure	1	—	0	0
Insufficient S+V test support	1	—	8	8

Table 20: E13 cross-database adaptation, **aggregated to unique subjects**. INCART recordings are grouped by the subject identifier in the WFDB header (11 recordings → 6 subjects); SVDB recordings are one per subject. Scores are per-subject macro-F1 over N/S/V with 95% bootstrap intervals over subjects, mean paired gain versus the frozen source, and subjects improved. With $n = 6$ the INCART rows are descriptive rather than inferential.

DB	Method	macro-F1	95% CI	Δ	Δ CI	Impr.
	Frozen source	0.459	$[0.372, 0.548]$	—	—	—
INCART	Head + post-pool	0.595	$[0.567, 0.621]$	+0.137	$[+0.058, +0.215]$	6/6
$n = 6$ subj.	Enc. LoRA + raw	0.618	$[0.601, 0.639]$	+0.160	$[+0.082, +0.234]$	6/6
	Enc. LoRA, no replay	0.647	$[0.630, 0.666]$	+0.189	$[+0.110, +0.264]$	6/6
	Frozen source	0.384	$[0.335, 0.434]$	—	—	—
SVDB	Head + post-pool	0.580	$[0.543, 0.618]$	+0.196	$[+0.152, +0.241]$	45/46
$n = 46$ subj.	Enc. LoRA + raw	0.610	$[0.566, 0.653]$	+0.226	$[+0.179, +0.274]$	46/46
	Enc. LoRA, no replay	0.626	$[0.578, 0.674]$	+0.242	$[+0.190, +0.297]$	45/46

Source-domain macro-F1 change on DS1 N/S/V: INCART -0.039 / -0.045 / -0.115 and SVDB -0.066 / -0.068 / -0.137 for head-only replay, encoder LoRA with raw replay, and encoder LoRA without replay respectively. Dropping replay buys the largest target gain at the largest retention cost.

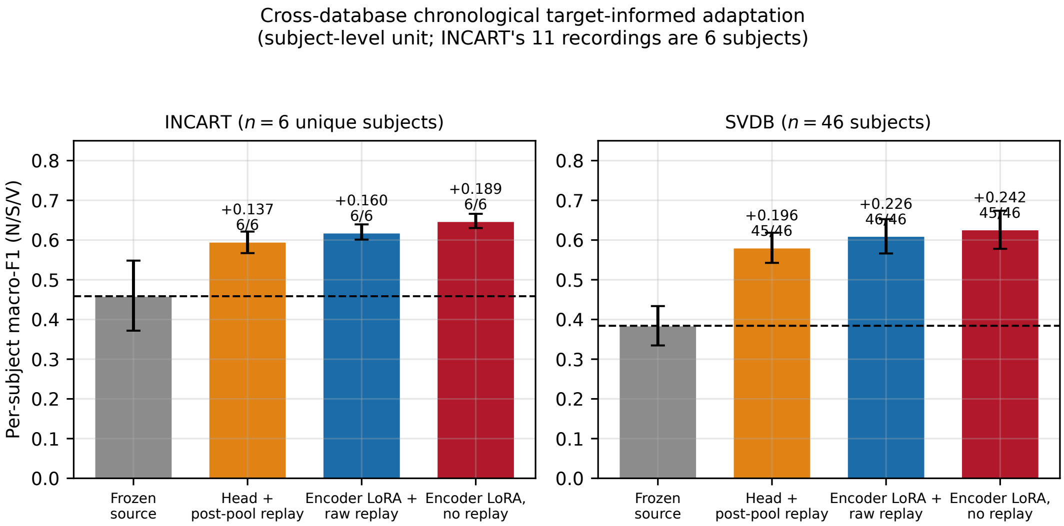

Figure 10: Subject-level cross-database adaptation on INCART and SVDB. INCART recordings are grouped by the subject identifier in the WFDB header, giving six unique subjects from 11 recordings; SVDB recordings are one per subject. Every target-informed method improves per-subject macro-F1 (N/S/V) over the frozen source (dashed line); annotations give the mean paired gain and the fraction of subjects improved, and error bars are 95% bootstrap intervals over subjects. Encoder LoRA has a higher mean than head-only replay on both external subsets; no paired direct-comparison interval is reported, so this is not a claim of statistical superiority. With $n = 6$ the INCART panel is descriptive rather than inferential.

5.10 RQ7 — Labeling burden and class coverage

Takeaway: around 100 labelled beats recover roughly 80% of the cohort-mean adaptation gain, but *which* labels matter more than how many — coverage of the patient's minority class roughly doubles per-patient recovery (secondary evidence).

Table 21 and Fig. 11 give the label-budget curve for A4 on the primary cohort under class-aware-oracle and repeated-random selection at $K \in \{25, 50, 100, 200, 500\}$, with patient-bootstrap bands. **The class-aware policy is a deliberately non-deployable oracle: it uses knowledge of which classes later appear in the patient's test segment to choose which beats to label, and therefore estimates an upper bound on the value of class coverage rather than a policy a device could execute.** The random policy is the deployable comparator; where the two differ, the gap is the headroom a realisable selector would be competing for. Measured on *cohort means*, the class-aware policy recovers 0.58 of the 500-label gain at 25 labels, 0.72 at 50, 0.83 at 100, and 0.93 at 200; random selection recovers 0.54/0.64/0.78/0.89.

The cohort mean overstates what an individual patient gets. The cohort figure is a ratio of mean gains. The quantity a deployment actually cares about is the per-patient recovery ratio $R_K = (M_K - M_0)/(M_{500} - M_0)$, and averaging *those* gives a noticeably weaker picture (Table 22). Under class-aware selection the median patient recovers 0.27 at 25 labels, 0.61 at 50, 0.67 at 100, and 0.94 at 200 — and **only 38% of patients individually reach $R_K \geq 0.8$ at 100 labels**, rising to just 52% at 200. The per-patient *mean* R_K is unstable at small K (it goes negative) because patients whose 500-label gain is near zero have a near-zero denominator; we therefore report the median and the fraction above threshold rather than the mean.

The defensible claim is consequently the narrow one: *around 100 labelled beats recover roughly 80% of the cohort-mean adaptation gain under the tested policies.* It is **not** true that 100 labels give 80% of the gain for a typical individual patient, and we do not claim that 50 labels suffice.

Which labels matter more than how many. Splitting patients by whether the adaptation window happened to contain the arrhythmia class that later appears in the test window is the strongest signal in this experiment. At 100 class-aware labels, patients with that coverage recover $R_K = 0.82$ on average against 0.41 for those without — roughly double. Label *budget* is therefore the wrong lever to optimize on its own: a small annotated set that happens to include the patient's minority beats outperforms a larger one that does not, which argues for coverage-aware annotation prompting rather than simply asking a clinician for more labels.

Table 21: E12 label-budget curve for A4 on the 21-record patient-disjoint cohort (5 seeds). "Cohort recovery" is the fraction of the 500-label mean gain over A0 recovered at that budget. Per-patient recovery, which is materially weaker, is in Table 22.

Policy	Labels	macro-F1	95% CI	Cohort recovery
	25	0.745	$[0.672, 0.818]$	0.582
	50	0.764	$[0.692, 0.835]$	0.716
Class-aware oracle	100	0.780	$[0.705, 0.853]$	0.832
	200	0.793	$[0.717, 0.867]$	0.931
	500	0.802	$[0.733, 0.873]$	1.000
	25	0.740	$[0.669, 0.813]$	0.541
	50	0.754	$[0.681, 0.828]$	0.643
Repeated random	100	0.773	$[0.698, 0.848]$	0.784
	200	0.787	$[0.713, 0.858]$	0.885
	500	0.802	$[0.731, 0.875]$	1.000
Reference points, full 500 labels, chronological				
A0 frozen	500	0.666	$[0.563, 0.766]$	—
A1 head + max replay	500	0.786	$[0.712, 0.859]$	—
A5 LoRA r=2	500	0.809	$[0.734, 0.883]$	—

Table 22: Label-budget recovery on the 21-record patient-disjoint cohort (5 seeds). "Cohort" is the ratio of mean gains, the figure usually quoted; R_K is the per-patient ratio $(M_K - M_0)/(M_{500} - M_0)$. The per-patient *mean* is unstable at small K — patients whose 500-label gain is near zero give a near-zero denominator — so the median and the fraction clearing $R_K \geq 0.8$ are the meaningful summaries. The last two columns split patients by whether the adaptation window contained the arrhythmia class that later appears in the test window.

Policy	K	Cohort	Mean R_K	Median R_K	$R_K \geq 0.8$	R_K covered	R_K not
	25	0.582	−0.75	0.27	19%	−0.47	−0.94
	50	0.716	−0.03	0.61	38%	0.31	−0.25
Class-aware oracle	100	0.832	0.57	0.67	38%	**0.82**	0.41
	200	0.931	1.08	0.94	52%	0.66	1.37
	500	1.000	1.00	1.00	100%	1.00	1.00
	25	0.541	−0.56	0.29	24%	−2.61	−0.36
	50	0.643	0.02	0.38	29%	0.17	−0.02
Repeated random	100	0.784	0.53	0.66	29%	0.65	0.50
	200	0.885	1.01	0.84	52%	0.94	1.03
	500	1.000	1.00	1.00	100%	1.00	1.00

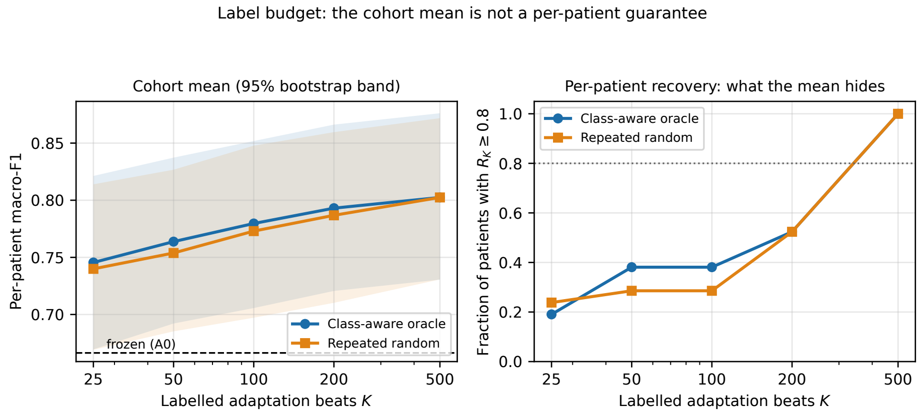

Figure 11: Label budget for A4 on the primary cohort, shown two ways because the two disagree. **Left:** cohort-mean macro-F1 with patient-bootstrap 95% bands — the curve usually reported, on which class-aware selection reaches ≈ 80% of the 500-label gain near 100 labels. **Right:** the fraction of *individual* patients whose recovery ratio R_K clears 0.8 (dotted line). That fraction is 0.38 at 100 labels and still only 0.52 at 200, so the left panel must not be read as a per-patient guarantee (Table 22).

5.11 Source-domain change

Takeaway: tens of raw replay exemplars keep source-domain loss small and bounded — encoder arms do not incur larger source-domain loss than head-only arms despite touching the encoder, and no source class collapses.

Across A1–A5 the source-domain macro-F1 change on the held-out DS1 N/S/V classes stays within $[-0.048, -0.027]$ (frozen reference 0.946), and the five-class change stays within ± 0.024. Encoder arms do not incur larger source-domain loss than head-only arms despite touching the encoder: tens of raw replay exemplars are enough to anchor it, and no source class collapses. Fig. 12 plots per-configuration target gain against source-domain change; the encoder arms sit in the high-gain, small-source-loss region rather than trading one for the other.

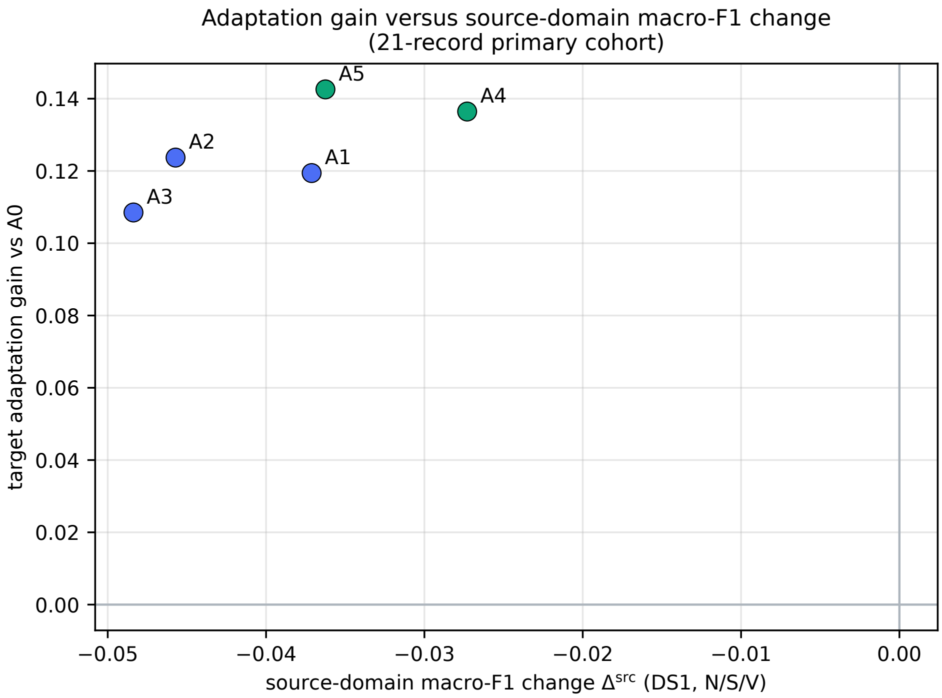

Figure 12: Target patient macro-F1 gain versus the signed source-domain macro-F1 change Δ^{src} (Eq. 1) for sweep configurations. Δ^{src} is negative when adaptation costs source performance, so points nearer the top-right retain more of the source while gaining more on the target. Target gains are not bought with a proportional source-domain loss; the encoder arms sit near the stability–plasticity Pareto edge. The axis is labelled source-domain macro-F1 change throughout; we avoid the word "forgetting" because the quantity measured is a change in DS1 score, not a mechanism.

6 Discussion

6.1 The replay-point cliff is not an optimization answer

The cliff measurement is deterministic: only post-pool replay is smaller than the raw input for this architecture. The primary results change its interpretation. Compression determines how many exemplars *fit*, but not how useful the remaining trainable subspace is. Maximum post-pool replay (A1) fixes the representation and buys volume; raw replay is expensive per item but leaves budget for attention-level updates. Across all 21 patients, the second allocation improves more patients (17/21 versus 12/21) and every pooled minority-class metric, even though it stores an order of magnitude fewer exemplars. The cliff tells you where you *may* replay; it does not tell you where you *should*.

6.2 Why encoder adaptation may help

The source model has strong V sensitivity but weak S sensitivity. S identification depends heavily on prematurity and RR context, yet the base architecture fuses only a two-dimensional RR projection *after* pooling. An encoder LoRA update can reshape morphology–context interactions upstream of the pool that a replacement linear head cannot reach. The F-class result is consistent: post-pool arms plateau near 0.39 F1 while encoder arms reach 0.58–0.60. This remains a mechanistic *hypothesis*. RQ5 is the test that speaks to it most directly, and it is weakly consistent with the hypothesis rather than confirmatory: starting from sources whose S sensitivity is already 0.78–0.88 *on their own DS1 test split*, encoder adaptation still matches head-only replay ($\Delta_{\mathrm{A4-A1}} = +0.0001$ and $+0.0024$). The adaptation gain persists under two alternative class-reweighted source-training objectives. This supports robustness to the choice of source objective, but does not establish robustness to a demonstrably stronger unseen-patient source model. Those margins are also small and carry no reported uncertainty, so they do not establish an ordering.

6.3 What the data support

Supported by the 21-record patient-disjoint, five-seed primary evidence:

- patient-specific, target-informed adaptation improves the primary cohort under the analytical 16 KiB budget;
- both encoder-LoRA arms significantly beat the frozen model at patient level (Holm $p \leq 0.008$), with large rank-biserial effects and CIs excluding zero — these are the *only* two of the six pre-specified comparisons that survive Holm correction;
- encoder arms improve more patients and give better pooled S and F performance than maximum head-only replay, with $\approx 12\times$ fewer exemplars, though the margin in improved-patient count narrows from six patients to two once a meaningful-change threshold is applied;
- the target-side benefit is attributable to encoder plasticity (B4−B2, $p = 0.005$, 16/21 patients); the replay contrast is positive in mean but not statistically supported (B4−B3, $p = 0.473$, median $\approx$ zero), and replay *volume* shows **no statistically supported target-side benefit under the tested selector, replay ratio, optimizer, and patient cohort** (B1−B6, 5 improved / 15 worsened);
- replay provides substantially better source retention than the tested zero-byte B-factor anchor under the evaluated configuration: at matched bytes, replay gives $+0.136$ target gain against the anchor's $+0.078$ while losing about a fifth as much source performance (-0.027 versus -0.136 macro-F1), and the anchor adds nothing once replay is present. Only this one scale-ambiguous regularizer was tested, so this does not speak to EWC, distillation, output consistency, or a $\|AB\|_F^2$ penalty (Section 5.4);
- about 100 labeled beats recover $\approx 80\%$ of the *cohort-mean* 500-label gain, and whether the adaptation window covers the patient's future arrhythmia class matters more than the label count (RQ7);
- the conclusions do not depend on saturating the budget: re-running the whole grid against a 15,360 B ceiling, i.e. holding back a 1 KiB firmware reserve, preserves every significant gain over the frozen model and keeps encoder adaptation among the top allocations. The *exact arm ordering is not preserved* — A1 rises to 0.793 while A5 falls to 0.792, putting A4 first — which is why we read rank-1 as the more robust operating point (Section 5.6).

Supported by secondary evidence, on reduced panels or other cohorts:

- (RQ4) within the measured 12-record Adam panel at ranks 1 and 2, raising the adaptation-state budget from 8 to 64 KiB produced no statistically detectable benefit; 14 of 18 fixed-configuration contrasts met the pre-specified ± 0.05 equivalence criterion, uncorrected for multiplicity;
- the adaptation gain survives adaptation from two class-reweighted DS1 sources, of five trained; the encoder-minus-head margins there are nonnegative, but they are small and carry no reported uncertainty (RQ5);

- chronological target-informed adaptation transfers to two external databases at *subject* level, with +0.14 to +0.24 mean gains on INCART (6/6 subjects) and SVDB (45–46/46); encoder LoRA has the higher mean on both external subsets, without a paired direct-comparison interval (RQ6).

Not supported:

- direct statistical superiority of A4/A5 over A1 on the primary cohort (the direct comparisons remain non-significant, and the reweighted-source and external margins, while consistently positive, are small);
- any optimizer ranking, or any rank-4/rank-8 statement: SGD is measured only at 8 KiB, SGD-with-momentum not at all, and 86 of 140 planned sweep cells are unrun;
- *inferential* external validity on INCART: the subject-level gain is positive with an interval excluding zero, but $n = 6$ subjects supports a descriptive reading only, and the cohort is a tractability-capped sample of 13 of 75 records rather than a principled selection;
- RR-architecture and CNN source baselines (three E11 variants still require new model graphs);
- any hardware runtime, SRAM, latency, or energy claim;
- a "50 labels are sufficient" claim, or any per-patient reading of the label curve: at 100 labels only 38% of patients individually reach 80% recovery, so the cohort-mean figure must not be read as a per-patient guarantee.

6.4 Embedded interpretation

Everything reported here is persistent-state accounting (Appendix B gives the packed record layouts, the memory-category table, and a qualitative discussion of the unmatched compute paths): trainable weights, gradients, optimizer moments, replay values, labels, and metadata, packed and asserted per configuration. It excludes forward and backward activations, stack and workspace, allocator overhead, and DMA buffers, all of which are peak-SRAM rather than persistent-state quantities and all of which are implementation- and toolchain-specific. The 16 KiB figure is therefore a statement about what a device must *retain* between adaptation steps, not about what it must have free at any instant; a hardware study would need to add the runtime terms.

The budget is also accompanied by a firmware implementation reserve. An arm consuming 16,383 of 16,384 bytes is not shippable, so we re-generated all configurations against an effective ceiling of 15,360 B and re-ran every arm at the reduced replay counts rather than reporting old numbers against new byte figures. That costs A1 49 exemplars and the encoder arms five each. It changes no significance conclusion, but it does reshuffle the arm ordering — A5 is the arm most sensitive to losing its last few replay samples — so the reserve replication is what identifies rank-1 as the safer engineering choice (Section 5.6).

There is one cost of the encoder allocation that the byte budget hides. Raw replay requires re-forwarding stored beats through the encoder and propagating gradients through the LoRA training path, whereas post-pool replay enters the graph after the encoder and bypasses it. Encoder adaptation therefore has substantially greater compute and energy cost per adaptation step, but **this study does not measure or reliably estimate its magnitude** and we quote no ratio (Appendix B explains why the earlier analytical figure was withdrawn). The arms are matched in bytes and not in compute, which on a duty-cycled device is an energy difference a deployment study would have to quantify before choosing between them.

The external result adds a nuance to the frozen-replay story: under strong domain shift, the no-replay encoder LoRA achieves the *largest* target gain but the *most* source-domain change. Replay is therefore best understood not as a pure accuracy lever but as a stability regulator whose value grows with the cost of source-domain loss.

The regularization comparison sharpens that reading. A parameter-space anchor and a replay buffer are both stability mechanisms, but they are not interchangeable: the anchor constrains *how far the weights may move*, whereas replay constrains *what the function must still do.* On a shift that is a change of patient rather than a change of task, the second constraint is the useful one, and the measurements bear that out — at equal persistent bytes, replay incurs roughly one fifth of the anchor's source-domain loss while also adapting further. We hold this to its tested scope: *replay provides substantially better source retention than the tested zero-byte B-factor anchor under the evaluated configuration.* It is one regularizer, in a scale-ambiguous form, so it does not license the broader claim that a budget should never be spent on regularization — restricted EWC, source-logit distillation, output-consistency regularization, and a $\|AB\|_F^2$ penalty are all untested here and could behave differently.

6.5 Clinical interpretation

The method is not an autonomous diagnostic system. It assumes supervised labels for an early patient segment and evaluates heartbeat classification on curated research databases. The strong patient heterogeneity—a few records personalize dramatically, most gain modestly, and a minority regress—is itself a clinical finding: personalization is unevenly beneficial, and a deployment would need a per-patient safeguard rather than a blanket "adaptation always helps" assumption. The results are algorithmic evidence about memory allocation, not clinical validation.

7 Limitations and Remaining Work

We state the current status of each limitation explicitly, including the two that were previously hard submission gates.

The evaluation cohort had to be corrected for a subject leak, and this matters. The de Chazal DS1/DS2 split is record-disjoint but not subject-disjoint: records 201 (DS1) and 202 (DS2) are the same person, per record 202's own WFDB header. Any result computed on the full 22-record DS2 set includes one patient the source model was trained on. We audited all 48 records, found this to be the only such pair, and excluded record 202. The correction is not cosmetic — it removed the statistical significance of the A4–A3 contrast (Holm p 0.047 $\rightarrow$ 0.075), leaving only the two comparisons against the frozen model surviving correction. Published results on this split that do not exclude 202 are not strictly patient-disjoint and may be optimistically biased by the same mechanism, and we recommend the check be run as standard.

Robustness to source-training objective (RQ5) — partially addressed. Five architecture-identical DS1 source variants (cross-entropy, effective-number weighting, focal $\gamma \in \{1, 2, 3\}$) are trained with five seeds and reach S sensitivity 0.78–0.88 on their own DS1 test split, a figure that motivated the concern. The adaptation gain persists under two alternative class-reweighted source-training objectives (Table 18). This supports robustness to the choice of source *objective*, but does **not** establish robustness to a demonstrably stronger unseen-patient source model. Three gaps remain. Adaptation was re-run from two of the five trained variants, so the robustness claim covers two sources rather than five. The three RR-placement and CNN variants (RR-conditioned token embedding, RR token before attention, and a lightweight inter-patient CNN) require new model graphs and were never trained. The encoder-minus-head margins from the reweighted checkpoints, while nonnegative, are small and carry no reported uncertainty, so RQ5 does not independently support an encoder-over-head ordering. These checkpoints are also not *stronger* sources — on intra-patient DS1 S sensitivity they are below the cross-entropy baseline — so RQ5 tests robustness to the source objective, not adaptation from a stronger start.

Cross-database transfer (RQ6) — closed at subject level, with a cohort caveat. INCART recordings were grouped by the subject identifier in the WFDB header and the saved cells re-aggregated to one score per subject, so the external analysis no longer treats repeated recordings from one person as independent. The gain survives the correction: +0.160 macro-F1 for encoder LoRA over 6 subjects with a bootstrap interval excluding zero and 6/6 subjects improved, against +0.226 over 46 subjects on SVDB. Two limits remain. With $n = 6$ the INCART result is descriptive rather than inferential, and the INCART cohort is a tractability-capped convenience sample — only the first 13 of 75 records were attempted — so it is not a principled sample of the database. Both external cohorts now have itemized inclusion and exclusion manifests (Table 19) in which every configured recording carries exactly one deterministic disposition. Source-only cross-database generalization remains weak; it is the target-informed adaptation that is positive.

The budget sweep is partial, and named accordingly (RQ4). Of 140 planned budget $\times$ optimizer $\times$ rank $\times$ replay-location configurations, 49 are measured, 5 are analytically infeasible, and 86 are unrun (Table 23). The measured surface is complete only for Adam at ranks 1–2, and only on a reduced 12-record, 3-seed panel. This supports only the narrow claim we make—that *within the measured 12-record, three-seed Adam rank-1/rank-2 panel, increasing the budget from 8 to 64 KiB produced no statistically detectable benefit*, tested by paired contrasts and TOST equivalence rather than by interval overlap—and supports nothing about optimizer choice, higher ranks, or budgets outside that panel. Numbers from this panel are never compared against primary-cohort arms. Following the completed scope, we describe E10 as an *Adam budget sweep for rank-1 and rank-2 adaptation*, not as the full optimizer grid it was originally designed to be.

Remaining limitations. (i) The primary-cohort direct A4/A5-vs-A1 comparison remains non-significant ($n = 21$), so a definitive "plasticity beats volume" claim is not licensed; a larger patient cohort is the most plausible route to settling it, and the cohort is now one record smaller than the protocol's nominal size. (ii) Persistent-state totals are packed and asserted per cell, and the whole grid was replicated against a 1 KiB firmware implementation reserve with no change in conclusions (Section 5.6); a byte-exact MCU port would still add allocator overhead not modeled here. (iii) Peak SRAM, update latency, update energy, and Flash write volume are all unmeasured, and no hardware training loop was executed. **Training compute, update latency, cycles, and energy are unmeasured, and no numerical compute ratio is reported**, because the encoder-LoRA backward cost requires layer-level profiling or a complete reverse-mode derivation that this study does not provide (Appendix B). (iv) The regularization family is represented here by *one inexpensive, memory-matched B-factor anchor* (Section 5.4), chosen because full-model FP32 EWC costs 53,144 B and is infeasible at this budget. Note it anchors the B factor, not the functional update BA. Restricted EWC, distillation from the frozen model, output-consistency regularization, and functional-delta ($\|AB\|_F^2$) regularization all remain untested, so this is a comparison against one convenient member of the family and not against the family. (v) MIT–BIH is small and old, and Q/paced beats are unsupported in the primary cohort. (vi) The reproducibility audit (Table 24) returns 11 PASS, 0 FAIL, and one WARN: configuration values are fixed in a versioned config file, but no automated provenance proves the hyperparameters were chosen without access to final test results. We report the WARN rather than suppressing it; the mitigation is that hyperparameters are shared across all arms, so any residual selection effect applies to every arm equally and cannot by itself produce the between-arm ordering. (vii) The method assumes supervised labels for an early patient segment and has had no prospective clinical evaluation.

Framing rule. Under a fixed analytical adaptation-state budget, patient-specific adaptation of the tiny ECG transformer exhibits a *replay–plasticity allocation frontier*. Encoder low-rank adaptation with limited raw replay significantly improves over the frozen model, produces stronger pooled minority-class performance, and yields more threshold-free positive patient changes than maximum-volume post-pool replay. However, the direct average difference from maximum-volume head-only replay is **not statistically significant**. The evidence therefore supports co-design and several competitive operating points rather than a universal winner, and no sentence in this paper should be read as establishing superiority over A1.

8 Conclusion

This software study establishes an algorithmic replay–plasticity trade-off for memory-constrained, patient-specific adaptation of a tiny ECG transformer, under a hardware-motivated 16 KiB persistent-state constraint. The architecture exhibits a deterministic replay-point cliff—tokens and feed-forward tensors are several times larger than the raw beat, and only the post-pool representation is storage-compressive—which makes maximum-volume post-pool replay the naive budget-optimal design. Over all 21 eligible MIT–BIH DS2 records and five seeds, that naive choice is not established as uniquely optimal: rank-1 and rank-2 encoder low-rank adapters with only tens of raw replay samples raise per-patient macro-F1 from 0.666 to 0.803 and 0.809, improve 17 of 21 patients (versus 12 of 21 for maximum head-only replay), and are both significant against the frozen model (Holm $p \leq 0.008$) with large effect sizes. Paired ablations attribute the target gain to encoder plasticity ($p = 0.005$) but leave replay's direct contribution statistically unsupported ($p = 0.473$), while replay provides substantially better source retention than the tested zero-byte B-factor anchor under the evaluated configuration; within the measured 12-record Adam rank-1/rank-2 panel, increasing the adaptation-state budget from 8 to 64 KiB produced no statistically detectable benefit, with 14 of 18 fixed-configuration contrasts meeting the pre-specified ± 0.05 equivalence criterion; and source-domain change stays bounded with no class collapse.

The direct comparison between encoder adaptation and maximum head-only replay is not statistically significant on the primary cohort, so the honest conclusion is a *replay–plasticity allocation frontier* rather than "depth beats volume." Encoder plasticity is more consistent and more replay-efficient at the same budget, and it carries a higher compute cost; it is not proven strictly superior to maximum replay, which is simply not established as uniquely optimal. The frontier holds up under stress, with one caveat we state rather than bury: the adaptation gain survives when adaptation starts from two class-reweighted DS1 source checkpoints (of five trained, with S sensitivity 0.78–0.88), though the encoder-minus-head margins there are small and carry no reported uncertainty; and the gain reproduces at subject level on two external databases, where INCART (6/6 subjects) and SVDB (45–46/46) show +0.14 to +0.24 mean gains, with encoder LoRA again showing the higher mean than head-only replay. Holding back a 1 KiB firmware reserve preserves every gain over the frozen model but reshuffles the arm ordering, which is what makes rank-1 the more robust operating point. What remains is well defined: adaptation from the three remaining trained source variants, three RR-architecture / CNN variants that were never trained, the unrun optimizer and higher-rank cells of the budget grid, a larger external cohort than six INCART subjects, and—for the strong "plasticity beats volume" claim—a larger patient cohort that could resolve the still-non-significant direct comparison. We make no claim of microcontroller deployment; the budget is analytical persistent state, and peak SRAM, latency, and energy are left to future hardware-in-the-loop work.

A Experiment Matrix and Reproducibility

A.1 Experiment ledger

The package comprises twelve experiments (E6–E17), summarized below with their status in this manuscript. Cohort and seed count are stated because they differ between the primary and secondary experiments.

Table 23: Experiment status. "Primary" denotes the all-DS2 cohort of 21 patient-disjoint records with five seeds; secondary experiments use reduced panels or other databases and are never pooled with primary results.

ID	Status	Content
E6	complete	Complete persistent-state accounting for 8/16/32/64 KiB, Adam/SGD, INT8 replay; every arm asserted $\leq$ 16,384 B per cell.
E7	complete (primary)	All-DS2 primary evaluation: 21 patient-disjoint records $\times$ 5 seeds, gap_beats = 1, 630 cells. Record 202 excluded as non-disjoint from DS1 (Section 4.2).
E8	complete (primary)	Six pre-specified paired comparisons with Holm correction, patient-bootstrap CIs, and rank-biserial effects.
E9	complete (primary)	Controlled replay-versus-plasticity matrix B0–B7 (B5 infeasible at 16 KiB) plus fixed-count and fixed-scope controls.
E10	partial (secondary)	*Adam budget sweep for rank-1 and rank-2 adaptation.* Designed as a full budget $\times$ optimizer $\times$ rank grid; executed on a reduced 12-record, 3-seed panel (864 cells) covering Adam at ranks 1–2 only. Of 140 planned configurations, 49 are measured, 5 analytically infeasible, 86 unrun (SGD at 8 KiB only, SGD-with-momentum absent, ranks 4–8 essentially unmeasured). Supports only the budget-saturation claim of Section 5.5; no optimizer or higher-rank claim is made.
E11	partial	Five class-reweighted DS1 source checkpoints (CE, effective-number, focal γ1/2/3) *trained* and measured for source quality with 5 seeds. Adaptation was re-run from **two** of them (effective-number, focal γ=2), so the robustness claim covers two sources, not five. Three RR-architecture/CNN variants `not_implemented_not_run`.
E12	complete	Label-budget curve (class-aware and random) with class-coverage and help/harm counts.
E13	complete (subject level)	Cross-database INCART and SVDB: frozen, head-only replay, encoder-LoRA replay, and no-replay encoder-LoRA. The saved cells retain the external record name, so results were re-aggregated to unique subjects (INCART 11 recordings $\rightarrow$ 6 subjects; SVDB one recording per subject) and bootstrapped over subjects. Itemized inclusion/exclusion manifests accompany both databases.

ID	Status	Content
E14	complete	Leakage/reproducibility audit (PASS 11 / FIXED 1 / ASSESSED 1 / WARN 1 / FAIL 0), split manifests, source checkpoint hash, config package.
E15	complete (primary)	Memory-matched regularization baseline: rank-1 encoder LoRA with an L2 source anchor $\lambda \sum_m \|B_m\|_F^2$ on the B factor of each attention projection, with (R2) and without (R1) raw replay, 21 records × 5 seeds, 220 cells. The anchor costs zero persistent bytes because the frozen base weights are the anchor, so R1 is byte-comparable with B3 and R2 with A4. Penalising B rather than the product AB is scale-ambiguous; reported as a low-cost baseline, not the strongest regularizer.
E16	complete (primary)	Implementation-reserve replication: the full A0–A5 grid re-run against an effective ceiling of $16{,}384 - 1{,}024 = 15{,}360$ B, which reduces replay counts (A1 705 → 656, A4 62 → 57, A5 52 → 47). 630 cells. Every gain over the frozen model remains significant and encoder adaptation stays among the top allocations; the exact ordering shifts (A5 loses more than A4), making rank-1 the more robust operating point.
E17	complete (sensitivity)	Boundary-leakage sensitivity rerun. The beat store was rebuilt *split-first*: the raw signal is cut at each record's chronological boundary, a 720-sample (2 s) guard is discarded from each side against a measured 120-sample filter influence reach, and the two pieces are filtered independently, so no adaptation-side sample can depend on a test-side raw sample. Arms A0, A1, A4, A5 over the same 21 records × 5 seeds, 420 cells, 497 adaptation beats per record. Every arm mean moves by ≤ 0.007 macro-F1; both encoder-versus-frozen contrasts stay significant ($p = 0.0012$) and both encoder-versus-A1 contrasts stay non-significant (Section 5.7).

A.2 Leakage and reproducibility audit

The audit driver executes twelve automated checks against the released configuration and the cached beat store, plus two checks added after the original audit missed them: subject disjointness across the DS1/DS2 split, and filter support across the chronological boundary. The summary is **PASS 11 / FIXED 1 / ASSESSED 1 / WARN 1 / FAIL 0**. Both added checks initially failed. The subject-disjointness failure is *resolved* by excluding record 202, hence FIXED. The filter-support failure is *characterised rather than resolved*: the primary pipeline still filters whole records, and E17 (Section 5.7) shows only that removing the dependency changes no conclusion — hence ASSESSED, not FIXED. The warning is reproduced and explained rather than suppressed. We record the two misses explicitly because an audit that only ever reports PASS is not evidence of anything.

Table 24: Audit outcomes and their consequences for the claims in this paper.

Status	Check	Evidence and consequence
PASS	DS1/DS2 record-list disjoint	Record-list intersection empty.
FIXED	**DS1/DS2 *subject* disjoint**	The record lists are disjoint, but records 201 (DS1) and 202 (DS2) are the same subject per 202's WFDB header. All 48 records were audited for this failure mode and 201/202 is the only cross-split pair. Record 202 is now excluded, giving 21 genuinely disjoint evaluation records. This check did not exist in the original audit.
PASS	Beat cache coverage	All configured DS1/DS2 records present in the cache; no silent record drops.
PASS	Cohort selection is outcome-blind	The primary panel equals the configured DS2 list; later-test arrhythmia burden is not used for inclusion.
PASS	Chronological adapt/test split	No record violates ordering; per-record index manifest released.
PASS	No window crosses the boundary	The guard beat prevents centred-window overlap; zero overlaps detected.
ASSESSED	**Whole-record filter dependency**	The original audit checked beat-window overlap but not filter support. Baseline-wander median filters and the zero-phase low-pass are applied to the whole record before the split, so adaptation-side samples within ≈ 120 samples of the boundary depend on test-segment raw samples (no test label, beat, normalization statistic, or model-selection signal is involved). Measured influence reach: 84 samples at 10^{-3} relative tolerance, 120 at 10^{-5}. **The primary pipeline retains this dependency**; E17 (Section 5.7) removes it and shows that no inferential conclusion changes. This check did not exist in the original audit.
PASS	RR normalization uses DS1 only	Computed statistics match the stored source statistics to eight decimals.
PASS	No future target statistics	Target RR normalization is called with adaptation-side indices only.
PASS	Early stopping is test-blind	Validation slices are drawn from the adaptation+replay union.
PASS	Per-patient reset	Each cell rebuilds from the frozen backbone; no adapted model carries over.

Status	Check	Evidence and consequence
PASS	Source checkpoint pinned	SHA-256 of the DS1 backbone recorded and released.
PASS	Seed count	Primary statistics use seeds $\{42, \ldots, 46\}$.
WARN	Hyperparameter selection is final-test independent	Values are fixed in a versioned config file, but no automated provenance proves they were chosen without final-test access. Consequence: a residual selection effect cannot be excluded. Mitigation: the same hyperparameters are used for every arm, so the effect applies uniformly and cannot by itself generate the between-arm ordering.

A.3 Runtime environment

Results were produced with TensorFlow 2.21.0 on a single NVIDIA RTX 3050 (6 GiB), seeds $\{42, 43, 44, 45, 46\}$, over the `all_ds2_primary` cohort (21 patient-disjoint records). Derived CSVs, publication figures, and LaTeX tables are regenerated from saved run reports by a single driver, so each table and figure in this manuscript is reproducible from the released configurations.

A.4 Extended ledgers

Tables 17 (source-baseline ledger) and 20 (cross-database ledger) are reproduced in Section 5. The label-budget ledger (Table 21) additionally reports the `secondary_informative_panel` single-seed measurements, which are retained only as secondary evidence and are not used for any headline claim.

B Persistent-State Serialization and Compute Scope

This appendix makes every byte in the budget claim reproducible and separates what the budget covers from what it does not.

B.1 What the 16 KiB covers

The 16 KiB constraint applies to persistent adaptation state, not to peak training SRAM. Table 25 lists every memory category and its status. The included categories are all quantities a device must retain between adaptation steps; the excluded categories are transient or implementation specific, and none of them is measured in this work.

Table 25: Memory categories and their treatment. "Replication only" marks the implementation reserve, which is held back in the reserve replication of Section 5.6 but *not* in the primary configuration. No sentence in this paper claims that total training SRAM is 16 KiB.

Memory category	In the 16 KiB?	Persistent?	Status
Trainable weights	yes	yes	analytical
Gradients	yes	assumption-dependent	analytical
Optimizer moments	yes	yes	analytical
Replay values	yes	yes	analytical
Replay labels	yes	yes	analytical
Buffer and quantization metadata	yes	yes	analytical
Alignment padding	yes	yes	analytical
Implementation reserve	replication only	yes	analytical
Forward activations	no	no	unmeasured
Backward activations	no	no	unmeasured
Stack and workspace	no	no	unmeasured
DMA and allocator overhead	no	no	unmeasured
Flash write endurance	no	—	unmeasured

Gradients are marked assumption-dependent because an implementation that applies updates immediately per batch need not retain a full gradient vector across steps. We include them so the accounting is conservative.

B.2 Packed replay records

Replay entries are serialized as fixed-stride packed records. With INT8 replay values, the raw record is

```
typedef struct __attribute__((packed)) {
    int8_t  ecg[198];   /* R-peak-centred beat, INT8          */
    int8_t  pre_rr;     /* normalised previous RR interval    */
    int8_t  post_rr;    /* normalised following RR interval   */
    uint8_t label;      /* AAMI class 0..4                    */
    uint8_t valid;      /* slot occupancy flag                */
    uint8_t class_id;   /* selector bookkeeping               */
} replay_raw_t;         /* 203 B packed                        */
```

and the post-pool record is

```
typedef struct __attribute__((packed)) {
    int8_t  pooled[16]; /* post-pool embedding, INT8          */
    int8_t  pre_rr;
    int8_t  post_rr;
    uint8_t label;
    uint8_t valid;
    uint8_t class_id;
} replay_postpool_t;    /* 21 B packed                         */
```

Both structures are all-`int8`/`uint8`, so the packed size, the natural ABI size, and the 1-byte-aligned stride coincide: 203 B and 21 B respectively. Under 4-byte alignment the strides become 204 B and 24 B, and under 8-byte alignment 208 B and 24 B; we report the 1-byte figures and account for the padding explicitly rather than assuming a favorable ABI.

Beyond the per-record fields, each buffer carries fixed metadata: a 4-byte write index, a 4-byte occupancy count, 16 bytes of reservoir/RNG state, one 4-byte counter per AAMI class (20 B), and per-quantized-tensor scale and zero point (4 B and 1 B each). This totals 49 B for the post-pool buffer, which quantizes one tensor, and 54 B for the raw buffer, which quantizes two (the beat and the RR pair). Table 3 reports the resulting totals.

B.3 Implementation reserve

A configuration that consumes 16,383 of 16,384 bytes is not deployable: it leaves nothing for a firmware version field, a checksum, or a future field addition. A deployable rule would hold back a reserve,

$$B_{\text{used}} \;\leq\; B_{\text{budget}} - B_{\text{reserve}},$$

which at $B_{\text{reserve}} = 1024\,\text{B}$ lowers the effective ceiling to 15,360 B.

We are explicit about how this relates to the results, because the two are easy to conflate. **The primary analysis does *not* apply a reserve.** Its configurations are generated against the full 16,384 B ceiling, because that is what the pre-specified analysis plan fixed, and changing the byte rule after the fact would forfeit the pre-specification. The reserve is instead reported as a *separate replication* (Section 5.6, Table 14): the entire A0–A5 grid re-run at the 15,360 B ceiling with regenerated replay counts, which preserves every significant gain over the frozen model but *not* the exact arm ordering — A1 rises to 0.793 and A5 falls to 0.792, so A4 leads under the reserve where A5 leads without it. Table 3 lists both count sets side by side so no arm's performance is ever quoted against a byte figure it was not measured under. A deployment should use the reserved counts; the primary table documents the pre-specified configuration.

B.4 Compute is not matched across arms

The arms are matched on persistent bytes but *not* on compute, and the direction of the gap is unambiguous even though this paper does not quantify it. Raw replay requires re-forwarding stored beats through the encoder and propagating gradients through the LoRA training path, whereas post-pool replay enters the graph after the encoder and bypasses it entirely. Encoder adaptation therefore carries a substantially greater compute and energy cost per adaptation step than head-only or post-pool replay.

We do not report a numeric ratio. An earlier version of this appendix gave analytical MAC totals and a fixed multiplier, charging backward cost as a multiple of the forward cost of the trainable path only. That convention is defensible for the head and post-pool arms, whose trainable parameters sit after the frozen encoder, but it is not a safe basis for a headline number in the encoder-LoRA arms: reverse-mode differentiation must propagate activation gradients *through* frozen operations to reach LoRA factors inserted inside the encoder, so a backward pass cannot be priced from the trainable parameter count alone. Rather than publish a ratio whose derivation we cannot fully stand behind, we state the qualitative asymmetry and leave the magnitude to a measured study — a layer-by-layer forward and backward derivation, or a profiler trace on the target runtime — in the journal version.

This matters for a deployment decision and not for the allocation result. The memory-allocation contribution of this paper is an accounting of persistent *bytes*, and every comparison in it is byte-matched by construction; none of the conclusions depends on the compute figures withdrawn here. A reader choosing between these arms on a duty-cycled device should nevertheless treat encoder adaptation as the more expensive option and measure it before committing.

Declarations

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Ethics. This study used publicly available, de-identified physiological databases distributed through PhysioNet [2, 3, 19] and did not involve new human-subject recruitment or intervention. No ethical approval was required.

Data availability. All data are public: the MIT–BIH Arrhythmia Database, the St. Petersburg INCART 12-lead Arrhythmia Database, and the MIT–BIH Supraventricular Arrhythmia Database, all obtainable from PhysioNet. The DS1/DS2 record partitions, the external inclusion and exclusion manifests, the INCART record-to-subject map, and every derived summary CSV behind the tables and figures are released with the code at the repository below.

Code availability. The adaptation framework, the constrained configuration generator, the memory-accounting module, the statistical pipeline, and the figure and table generators are released together with the run configurations, the random seeds, and the environment specification needed to reproduce every number in this paper. Every published value is regenerated from released artifacts by a single driver (`scripts/reproduce_paper.py`); no result is transcribed by hand.

Repository	`https://github.com/subhajitroy005/BudgetCL-ECG`
Software release	`v1.0.0-arxiv` (annotated, immutable tag)

The release tag is the primary identifier: a manuscript cannot contain its own Git commit hash, because inserting the hash changes the commit. The exact commit for this release is recorded in `releases/release_manifest.json` inside the tagged tree, so `v1.0.0-arxiv` resolves to it unambiguously.

The DS1 source checkpoint these results adapt from is pinned by

```
SHA-256 a6d4eff14caa4404eb57dc5eb9ecfcb9e9d3f1a1bf907d6d147fb5611eb79fce
```

so a reproduction can verify it is starting from the same model. Raw PhysioNet recordings are not redistributed; the repository provides download scripts and itemized inclusion/exclusion manifests instead.

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Author contributions. S.R. conceived the study, implemented the framework and experiments, performed the analysis, and wrote the manuscript.

Use of AI assistance. An AI coding assistant was used for software implementation, for generating analysis and figure scripts, and for editorial assistance during manuscript preparation. All experimental design decisions, all claims, and all interpretations are the author's; every reported number is produced by the released code from saved run artifacts, and the author has verified the correspondence between the manuscript and those artifacts.

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